Exclusive Contracts in the Video Streaming Market *

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Abstract

In the video streaming market, streaming services can use exclusive contracts to differentiate their content offerings and soften competition, while studios may leverage these contracts to negotiate higher licensing fees against streaming services. I investigate who gains and who loses from such contracts. I develop and estimate a structural model that incorporates bilateral negotiations between streaming services and studios, streaming services setting subscription prices, and consumer demand for subscriptions and titles. The findings reveal that the effect of exclusive contracts varies significantly across firms. Streaming services that lack in-house content (like Hulu) gain from exclusive contracts, while those with extensive in-house content (like Netflix) see minimal or even negative effects. For studios, exclusive contracts benefit small studios with weak bargaining power but harm large ones with strong bargaining power. Consumers lose from exclusive contracts due to reduced title distribution and higher subscription prices.

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1 Introduction

Exclusive contracts are prevalent across markets, such as Apple's exclusive deal with AT&T for iPhone sales from 2007 to 2011, Costco's with Visa for credit card processing since 2015, and Spotify's with Joe Rogan for podcast distribution between 2020 and 2024. These agreements often result from competitive negotiations where firms offer favorable deals to counterparties to secure exclusive rights, consequently shaping product availability, competition, and consumer welfare.

In this paper, I examine who gains and who loses from exclusive contracts in the video streaming market, where consumers pay subscription fees to access content libraries offered by services like Netflix and Hulu.[1] Using detailed data from various sources, I conduct an in-depth analysis on this increasingly important market, which now accounts for over a quarter of television viewing time in the U.S. (Nielsen 2022). Exclusive contracts, where studios grant exclusive licensing rights for titles to streaming services, are prevalent here. For example, Warner Bros. licenses *Friends* exclusively to Netflix, and Hulu has exclusive rights to *Fargo* from FX Networks (Slate 2018, Variety 2014). Consequently, 86.7% of titles from third-party studios are available on only one streaming service as of 2022. While such exclusive distribution can occur without exclusive contracts, these contracts are often used to ensure exclusive distribution.

Exclusive contracts are prevalent in the video streaming market, potentially driven by two forces that are common to many other markets. The first is differentiation: streaming services can use them to license unique content, differentiate themselves from competitors, and improve profitability. A second force, highlighted in this paper, is bargaining. Studios can use exclusive contracts as a bargaining tool, committing to exclusive distribution to play streaming services off against each other and negotiate favorable deals. This practice results in hefty license fees, such as Netflix's $100 million annual fee for the exclusive rights to *Friends* (Slate 2018).

The impact of exclusive contracts on studios and streaming services is theoretically ambiguous. The force of differentiation increases joint profits to be shared between them, benefiting both, while the force of bargaining shifts profits toward studios at the expense of services. This is further complicated by equilibrium effects: a contract between one pair of firms can affect the payoffs of other firms (Segal 1999). The direction and magnitude of equilibrium effects depend on the

[1] In this paper, "video streaming" specifically refers to Subscription Video on Demand (SVOD), excluding services like YouTube and Amazon Rent or Buy. I discuss market definition in greater detail in Section 2.

substitutability of both titles and subscriptions. The net impact of differentiation, bargaining, and equilibrium effects remains uncertain and can vary across studios and services.

Exclusive contracts harm consumers in the short term, when title production and streaming service participation remain fixed, since greater differentiation among streaming services forces consumers to subscribe to more services and pay higher prices to access desired content. However, in the long term, exclusive contracts may benefit consumers if they enhance the profitability of studios and streaming services, leading to increased content production or new service entry.

To quantify the net impact, I develop a structural model that captures both forces of differentiation and bargaining in the formation of exclusive contracts. The model incorporates bilateral contracting between streaming services and studios, subscription price setting by streaming services, and consumer demand for subscriptions and titles. I estimate the model using comprehensive data on viewership, subscription, and title distribution. Using model estimates, I quantify the welfare effects of exclusive contracts by simulating a counterfactual where they are absent, along with additional counterfactuals that account for adjustments in studios' content production and streaming service participation in response to profitability changes.

To capture the effect of differentiation, I integrate a model of demand for services subscriptions by households and titles by household members, along with a pricing model in which streaming services optimize payoffs. The demand model allows for consumer multi-homing across streaming services, heterogeneous viewer preferences, and varying influences of household members on subscription decisions. Together with the profit margins recovered from the pricing model, the demand model helps determine the expected profits streaming services can derive from specific titles through incremental subscriptions, which are split with studios in negotiations.

I estimate the demand model using subscription and viewership data. Subscription data detail market shares for all combinations of service subscriptions at the DMA-month level. The correlation between observed consumer multi-homing and changes in services' content libraries identifies the substitutability between service subscriptions (Gentzkow 2007). I observe title viewership at the weekly level, segmented by demographics such as age, gender, and race. Variance in viewership across demographic groups identifies diverse streaming preferences, while its covariance with subscription data helps identify the differential influences of household members. Lastly, I exploit geographical variations in tax rates on subscription demand to identify price elasticities, which inform the profit margins of subscriptions using the pricing model.

To capture the effect of bargaining, I develop a bilateral bargaining model in which streaming services negotiate with studios over inclusion in the distribution networks of titles—defined as the set of streaming services with licensing rights—and lump-sum license fees. During negotiations, a studio can threaten to replace a streaming service with an excluded alternative service to improve its bargaining leverage and demand higher fees (Ho and Lee 2019). This model highlights that studios with weaker bargaining power benefit more from improving their bargaining leverage through such threats of replacement, and therefore, have stronger exclusionary incentives.

Data on title distribution support this relationship: the "Big Five" studios (NBCUniversal, Paramount, Warner Bros., Walt Disney, and Sony), generally perceived to have stronger bargaining power, exhibit a 3.7 percentage point lower likelihood of exclusive distribution compared to small studios. Therefore, I estimate bargaining power by searching for parameters that best explain the observed distribution networks. This novel approach bypasses the need for data on privately contracted, lump-sum license fees.

The estimation results find that the mean own-price elasticity of service subscriptions is -1.45. The average household's willingness-to-pay for each title varies from less than \$0.01 to \$7.81. Small studios have weaker bargaining power (0.53) than the "Big Five" (0.82), leading to their stronger exclusionary incentives to improve bargaining leverage.

I apply the estimates to evaluate the impact of exclusive contracts by studying a counterfactual without them. This counterfactual reflects regulations in other markets where exclusivity raises concerns, such as the U.K.'s ban on exclusivity clauses in employment contracts for low-paid workers, which prevent them from working for multiple employers.

I find that the impact of exclusive contracts varies significantly across firms. On the streaming service side, small streaming services benefit substantially from exclusive contracts, with Hulu's payoff increasing by 110.1%, while large services see only modest or negative changes, with Netflix and Amazon Prime seeing an increase of 1.6% and a decrease of 5.3%, respectively. This disparity arises because small services lack in-house content, making them reliant on exclusive third-party titles for differentiation and competitiveness. In contrast, large services, with already substantial in-house content, see limited benefit from additional differentiation. This benefit can be offset by a negative equilibrium effect from increased competition from small services, which use exclusive third-party content to attract previous subscribers from these large services.

On the studio side, the "Big Five" see a 5.7% loss from exclusive contracts, while small studios experience an 8.1% gain. This contrast is driven by two countervailing effects. First, both groups face a negative equilibrium effect: exclusive contracts differentiate streaming services and reduce their substitutability, which lessens their loss of subscribers when losing a title, and therefore, lowers their willingness to pay for titles. However, exclusive contracts also strengthen studios' bargaining leverage by enabling them to more effectively exert the threat of replacement against streaming services. In addition, the increased profitability of small services makes them credible threats that studios can leverage in negotiations with large services. With weaker bargaining power, small studios gain more from this improved leverage, leading to their gains. In contrast, the "Big Five," with their already strong bargaining power, do not benefit as much from the improved leverage, resulting in an overall loss.

Finally, assuming fixed title production and streaming service participation, exclusive contracts lead to a \$24 decrease in the annual surplus of the average household. This loss results from both reduced title distribution and increased subscription prices. However, the gains for small studios and streaming services may stimulate both content production and streaming service entry in the long run. Analyzing additional counterfactuals that capture these margins, I find that they may mitigate or even reverse the short-run negative impacts of exclusive contracts on consumers.

Contribution and Related Literature This paper adds to the growing literature on exclusive contracts. Bork (1978) argues that exclusive contracts must be efficient to be adopted by firms, but subsequent research has identified both pro- and anti-competitive effects. On the positive side, they can mitigate contracting externalities (Segal 1999), encourage investment (Segal and Whinston 2000), and stimulate market entry (Lee 2013, Le 2023). Conversely, they can soften competition (Rey and Stiglitz 1995, Nurski and Verboven 2016, Sinkinson 2020), deter or foreclose market entry (Bernheim and Whinston 1998, Asker 2016), and raise competitors' costs (Subramanian, Raju and Zhang 2013). Most of these papers employ offer or bidding games. In contrast, this paper extends the framework to a broader bargaining game, thereby highlighting the role of bargaining in the rise of exclusive contracts. It aligns with recent theoretical insights from Chambolle and Molina (2023) and Abreu and Manea (2024), who study capacity restriction as bargaining leverage, but goes further by empirically investigating how bargaining power affects the formation of exclusive contracts and evaluating their welfare effects.

This paper contributes to the empirical literature on vertical contracting and bargaining. Observing detailed contractual terms, Mortimer (2007, 2008) and Ho, Ho and Mortimer (2012) study price discrimination, revenue sharing, and full-line forcing in the video rental market. Most research in this area (e.g., Draganska, Klapper and Villas-Boas 2010, Crawford and Yurukoglu 2012, Gowrisankaran, Nevo and Town 2015, Ho and Lee 2017, Crawford et al. 2018) adopts the "Nash-in-Nash" bargaining model, which assumes predetermined networks of agreements and requires negotiated prices as inputs. However, these conditions are not suitable for the video streaming market, where exclusive contracts are strategically used as a bargaining tool, and license fees are negotiated confidentially and paid as lump sums, making them neither observable nor inferable from optimal pricing conditions. Ho and Lee (2019) addresses the first limitation with the "Nash-in-Nash with Threat of Replacement" (NNTR) bargaining solution, allowing firms to use excluded counterparties as threats in bargaining—a feature also incorporated by Hristakeva (2022a,b), Beckert, Smith and Takahashi (2025), among others. To address both limitations, I extend NNTR to model endogenous network formation. Moreover, I identify bargaining power using variation in title distribution, as studios with weaker bargaining power have stronger exclusionary incentives. Compared with alternative network formation models (e.g., Liebman 2018, Ghili 2022), my approach circumvents the need for negotiated prices and does not assume fixed costs of bargaining to explain negotiation breakdowns, making it more widely applicable for studying vertical relations.

This paper also adds to the literature on the emerging streaming industry, which has examined aspects such as video streaming's competition with traditional TV (Malone et al. 2021, McManus et al. 2022) and content distribution strategies (Amaldoss, Du and Shin 2021, Berbeglia, Derdenger and Tayur 2023), and the welfare impact of new products in music streaming (Aguiar and Waldfogel 2018). Using new and comprehensive data on subscriptions, viewership, and title distribution, this study explores the competition and interactions across all major market stakeholders—studios, streaming services, and consumers.

Roadmap The remainder of the paper proceeds as follows. Section 2 introduces the video streaming market and its vertical contracting practices, and Section 3 develops a simple model reflecting these practices. Section 2.4 details the data and descriptive analysis. Section 5 outlines the empirical framework, with estimation details in Sections 6 and 7. Section 2.8 discusses counterfactual exercises and their implications. Section 2.9 concludes.

2 The Video Streaming Market

This paper focuses on the Subscription Video On Demand (SVOD) sector of the video streaming market, where users pay a subscription fee to access a digital content library. SVOD is a distinct market from transactional video on demand (TVOD) services like Amazon Rent or Buy, and advertising video on demand (AVOD) services such as YouTube, due to differences in revenue models and separately negotiated licensing agreements—even for services operated by the same company, such as Amazon Rent or Buy and Prime Video. Throughout this paper, I use "video streaming" and "SVOD" interchangeably.

This paper focuses on four leading streaming services—Netflix, Amazon Prime Video, Hulu, and Disney Plus—from March 2021 to February 2022. These four services dominated the market, claiming more than 75% of the total SVOD expenditure and viewing time in the U.S. during the period (Wall Street Journal 2021, Nielsen 2022). Other services were much smaller at the time, with HBO Max, the fifth largest, having about half the market share of Disney Plus, the fourth largest.

Streaming services offer two types of content: in-house and third-party titles. While streaming services have increased in-house productions significantly, third-party content remains their primary content source, accounting for nearly 60% of the top 2,000 titles in my data. Notably, 86.7% of these third-party titles are available on only one service. These unique content offerings lead to prevalent consumer multi-homing, with the average U.S. household subscribing to 1.7 of the top four services during the study period in my data.

Industry Practices of Vertical Contracting Licensing contract negotiations between studios and streaming services only involve third-party content.[2] Negotiations typically occur on a per-title basis, with most contracts outlining exclusive rights and *lump-sum* fees. There is no revenue sharing

[2]Streaming services often also have production contracts with studios, where studios produce in-house content for services and are compensated for production costs plus a premium. However, the streaming service retains full ownership of the content's intellectual property, so no licensing contracts are needed.

between studios and streaming services, even for advertising revenues.[3] Contracts usually last for a year, with rare opportunities for renegotiation.

Exclusive contracts have significantly affected the bargaining dynamics between streaming services and studios. Studios can commit to distribute to only one service, often through exclusive contracts, so they can go back and forth between services, using offers from one as leverage to squeeze another to pay more. This threat of exclusion and replacement pressures services to outbid their rivals, often resulting in substantial licensing agreements. For example, Netflix secured a $100 million annual exclusive agreement for *Friends* in 2019 (Slate 2018), while Amazon offered $90 million for *Crime 101* and made a multi-million dollar bid for *Sound of Freedom* in 2023 (Screen Rant 2023, Vulture 2023).

Exogenous Title Production In the main model, I assume title production is exogenous and unaffected by exclusive contracts due to studios' significant capacity constraints over the study period. Because of the demand surge following COVID-19, studios lacked the capacity to produce additional content within a 2–3 year time frame. However, I explore how exclusivity affects studio profits and analyze a counterfactual in Section 8.2 to study the potential welfare impact if studios adjust production in response to profitability changes.

3 A Simple Model of Bilateral Contracting

In this section, I present a simple model of licensing contract negotiations to illustrate the formation of exclusive contracts and their efficiency implications. Consider a market with a single-title studio, j, and two streaming services, k_1 and k_2, playing a two-stage game as illustrated in Figure 1:

- **Stage 1** (Bilateral Contracting): Studio j chooses a set of streaming services to reach licensing agreements with. Each contract is negotiated bilaterally and simultaneously, includes a *lump-sum* license fee τ, and can specify if the licensing right is exclusive.
- **Stage 2** (Downstream Competition): k_1 and k_2 compete downstream and realize sales profits. These profits are denoted as Π_1^e and Π_2^e when the title is only on each of the services, and as Π_1^{ne} and Π_2^{ne} when on both services.

[3] Netflix provides additional references on licensing payments: `https://ir.netflix.net/ir-overview/top-investor-questions/default.aspx`. Amazon pays some independent titles based on viewership but mainly uses lump-sum payments for popular titles, which account for most of its license fees and are the focus of this paper.

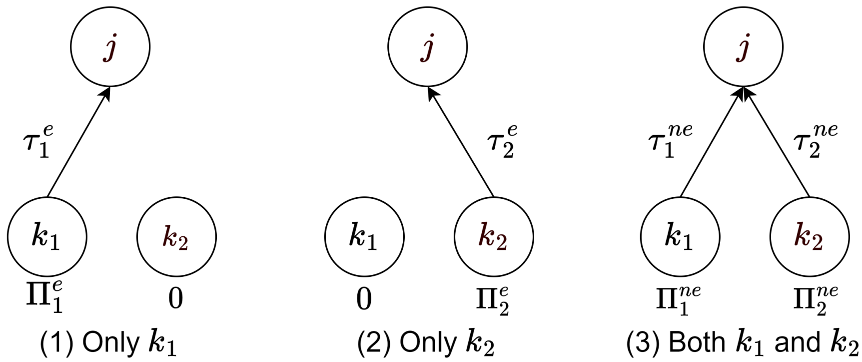


Figure 1: Market Outcomes Under Various Distribution Networks

I use pure-strategy weak perfect Bayesian equilibrium as the equilibrium concept. To refine equilibrium outcomes, I assume that either party may terminate a bilateral contract at will, following Ho and Lee (2019).

This simple model incorporates two key features. First, the studio can commit to exclusive distribution—often ensured by exclusive contracts in practice—to create licensing scarcity. Second, it can induce competition between streaming services by going back and forth between them, using the potential deal it can reach with one service as a bargaining threat against the other. With these two features, the model captures the studio's ability to leverage distribution restrictions to negotiate higher license fees, as highlighted in Section 2. In Appendix B.1, I present the extensive-form game and derive the corresponding equilibrium outcomes, which are detailed below.

License Fee Negotiations To capture these two key features, I adopt the Nash-in-Nash with Threat of Replacement (NNTR) bargaining solution from Ho and Lee (2019) to determine negotiated license fees. When studio j intends to only license to k_1, the NNTR solution of the license fee is

$$\tau_1^e = \arg\max_{\tau} (\tau)^b \left(\Pi_1^e - \tau\right)^{(1-b)} \text{ s.t. } \tau \geq \Pi_2^e, \tag{1}$$

where $b \in [0, 1]$ represents studio j's bargaining power against both services. τ_1^e maximizes the Nash product of gains-from-trade subject to a constraint, with τ as studio j's gain, and $\Pi_1^e - \tau$ as service k_1's. The constraint $\tau \geq \Pi_2^e$ highlights the studio's ability to improve bargaining leverage through commitment to exclusive distribution. With such commitment, the studio can create scarcity in licensing rights and induce k_1 and k_2 to compete for them. In practice, it achieves so by going back and forth between k_1 and k_2, using the offer from one as leverage when bargaining with the other. This approach ensures that k_1 pays no lower than the highest possible offer of Π_2^e from k_2, under which k_2 would break even.

I assume, without loss of generality, that $\Pi_1^e > \Pi_2^e$, meaning that k_1 can generate more profit than k_2 when the title is solely distributed to each service. Therefore, the studio must not enter into an agreement with k_2 alone because k_1, being more efficient, can always offer slightly more than k_2's maximum offer of Π_2^e while maintaining a positive payoff.

When studio j negotiates non-exclusive contracts with both k_1 and k_2, it has no excluded service as leverage. Therefore, the NNTR solution defaults to the "Nash-in-Nash" solution:

$$\tau_k^{ne} = \arg\max_{\tau} (\tau)^b (\Pi_k^{ne} - \tau)^{(1-b)}, k \in \{1, 2\}. \tag{2}$$

Above all, the total license fees for the studio under different *distribution network* $\mathcal{K}_j$, or the set of streaming services licensing the title, can be specified as

$$\Pi_j(\mathcal{K}_j) = \begin{cases} \max\{b\Pi_1^e, \Pi_2^e\}, & \mathcal{K}_j = \{k_1\} \\ b \cdot \Pi_1^{ne} + b \cdot \Pi_2^{ne}, & \mathcal{K}_j = \{k_1, k_2\} \end{cases}. \tag{3}$$

Exclusionary Incentives of Studios Building on Ho and Lee (2019), I analyze the conditions under which exclusive title distribution arises. As the title owner, the studio has the discretion to select its distribution network. Therefore, it will choose the exclusive distribution network, potentially facilitated through an exclusive contract, if and only if it generates a higher payoff than the non-exclusive network: $\max\{b\Pi_1^e, \Pi_2^e\} \geq b \cdot \Pi_1^{ne} + b \cdot \Pi_2^{ne}$.

This condition breaks down into two scenarios. The first is when the joint profit for the bargaining parties is higher if the title is distributed only to k_1,

$$\Pi_1^e \geq \Pi_1^{ne} + \Pi_2^{ne}. \tag{4}$$

This implies that exclusive contracts can improve contracting efficiency by enabling the studio to commit to a distribution network that maximizes the sum of bilateral surpluses from contracts—equal to joint profits in this example—between itself and the contracting services (Segal 1999).

This occurs when the exclusive title can differentiate k_1 enough to encourage k_2 subscribers to either switch to k_1 or multi-home on both services. The profitability of this approach depends on multiple factors, including consumers' willingness to multi-home. The easier it is to induce multi-homing, the more profitable it is for the studio to distribute only to k_1.

The other scenario arises when it is more profitable for the studio to commit to exclusive distribution and use k_2 as a threat to improve its bargaining leverage against k_1:

$$\underbrace{\Pi_2^e - b \cdot \Pi_1^{ne}}_{\substack{\text{Gain from Improved} \\ \text{Bargaining Leverage}}} \geq \underbrace{b \cdot \Pi_2^{ne}}_{\substack{\text{Loss from} \\ \text{Exclusion}}} \implies b \leq \frac{\Pi_2^e}{\Pi_1^{ne} + \Pi_2^{ne}}. \tag{5}$$

This strategy enables the studio to charge a license fee of at least Π_2^e from k_1, instead of $b \cdot \Pi_1^{ne}$ when the title is distributed to both services. Using this tactic, the studio effectively shifts the surplus division with k_1 in its favor.

Moreover, condition (5) highlights the role of bargaining in the rise of exclusive contracts, as studios with weaker bargaining power are more inclined to opt for exclusive distribution. Specifically, the studio's gain from improved bargaining leverage, $\Pi_2^e - b \cdot \Pi_1^{ne}$, decreases with its bargaining power b, while its loss from excluding k_2, $b \cdot \Pi_2^{ne}$, grows with b. In Appendix B.2, I show that the intuitions hold when incorporating heterogeneous bargaining parameters across streaming services.

Finally, in practice, there are multiple studios licensing various titles. It creates equilibrium effects not captured in this single-title model: a contract between one pair of studio and streaming service for one title affects the profitability of other contracts, such as Π_1^e and Π_2^{ne}. Such effects, known as "contracting externalities" (Segal 1999), shape the profits generated by various contracts, and therefore, affect their formation and influence which studios and streaming services gain or lose from exclusivity. In Section 5, I capture these equilibrium effects by incorporating all titles in the bargaining model, along with consumer demand and platform pricing models that account for interdependencies in the profitability of licensing contracts.

4 Data and Descriptive Evidence

4.1 Data

This paper utilizes novel data that fall into three main categories: (a) title viewership and characteristics, (b) title distribution and production, and (c) quantities and prices of service subscriptions. The data cover the top four streaming services—Netflix, Amazon Prime Video, Hulu, and Disney Plus—from March 2021 to February 2022. I briefly discuss each in turn.

Title Viewership and Characteristics I draw on two data sources for title viewership and characteristics. The primary source is Nielsen's title rating data, which reports weekly viewing time for each title by U.S. individuals aged two and above, with demographic breakdowns by age, gender, and race. Appendix C.1 provides more details on Nielsen's rating data. I supplement it with data from Reelgood, which details title characteristics such as release dates and the most related genre. Genres are aggregated into six categories: action, comedy, horror and thriller, kids, drama, and others.

The final sample includes the 2,028 most popular titles, including all titles with a maximum or average weekly viewership in the top quintile on at least one service. This sample accounts for 91% of the total viewership across platforms. Demographic breakdowns of viewership, visualized in Figure A.1, show significant differences in genre-specific time allocation across demographic groups, reflecting diverse streaming preferences.

Title Distribution and Production Reelgood provides data on title distribution across the top four streaming services, as well as information on titles' production studios and distributors. Of the sampled titles, 1,145 (59.0%) are third-party, which involves licensing contract negotiations between streaming services and studios.[4]

Data highlight the prevalence of exclusive distribution: 993 (86.7%) of the third-party titles are on only one streaming service.[5] In contrast, only 125 (10.9%) and 27 (2.4%) of them are distributed to two and three services, respectively.

[4]Titles produced by a streaming service or with a single service as its only global distributor are classified as in-house; all others are considered third-party. In the final sample, 88 titles have no production studio information.

[5]The raw data provide daily-level title distribution details. 222 titles (10.9%) in the final sample migrated across streaming services at least once during the study period. For simplification, a title's distribution network is defined by the unique combination of services where it has the longest period of distribution.

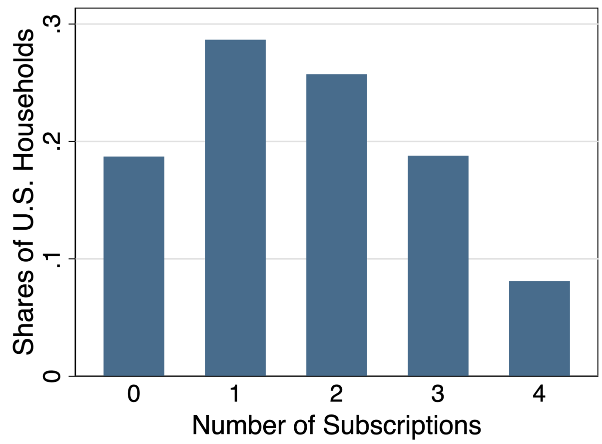


Figure 2: Distribution of the Number of Subscribed Services

Notes. This figure presents the monthly average share of households that subscribe to varying numbers of the top four streaming services within the study period and the top 30 DMAs, weighted by the population in each DMA.

Quantities and Prices of Service Subscriptions Service subscription quantities are sourced from the Nielsen Household Universe Estimates dataset, which provides monthly market shares of services at the DMA level for the 30 most populous DMAs.[6]

A key advantage of this dataset is its reporting of market shares for each bundle, or unique combination of streaming services. For example, it details the share of households in the Atlanta DMA subscribed to Netflix and Hulu but not to Amazon Prime and Disney Plus in June 2021. This unique feature enables an analysis of the prevalent consumer multi-homing behavior, where more than half of U.S. households subscribe to at least two of the top four streaming services, as described by Figure 2.

Subscription prices are sourced from streaming services' websites. In addition, consumers also pay taxes for subscriptions, which vary significantly nationwide. Most states impose sales taxes on streaming subscriptions, but some, such as California and Utah, do not. In addition, states and cities may levy unique taxes, such as Florida's communications services tax of up to 15% in some counties and Chicago's 9% amusement tax. Tax rates are collected from multiple sources: Bloomberg Tax identifies applicable tax categories by state and city, ZIP-code and month-level rates come from Thomson Reuters' Tax Data Systems, and state and local government websites cover special cases like Florida.

[6]Figure A.2 presents a map of these DMAs, which cover 55% of U.S. households in total.

Table 1: Distribution Networks of Titles

	(1) Exclusive Distribution		(2) Title-Service Distribution	
	Estimate	SE	Estimate	SE
"Big Five"	-0.038	0.019		
Disney's Studio			-0.236	0.034
Disney's Studio × Hulu			0.709	0.056
Other Controls	Title Characteristics		Streaming Service FE	
Observations	1,145		3,435	

Notes. The dependent variables in Column (1) and (2) are whether a title is distributed to only one service, and whether a title is distributed to a specific service, respectively. Title characteristics variables include release year, type (movie or show), and genre.

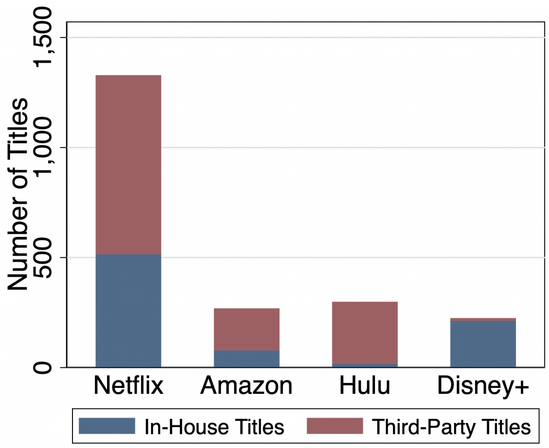
4.2 Descriptive Evidence on Title Distribution

Bargaining as an Exclusionary Incentive The contracting model predicts that studios with weaker bargaining power are more inclined to use exclusive distribution. To test this prediction, I compare the distribution strategies of the "Big Five" studios to smaller studios. The "Big Five," producing 46.1% of the sampled third-party titles, are expected to have stronger bargaining power due to their extensive bargaining experience and a century-long dominance in content production, which compels streaming services to make concessions to maintain long-term relationships and secure future licensing deals. Similarly, Crawford and Yurukoglu (2012) find comparable differences in bargaining power across content providers in the cable TV market using a Nash-in-Nash bargaining model.

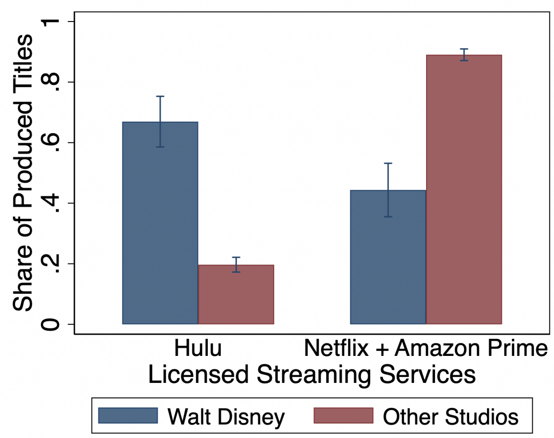
I find that 15.2% of titles from the "Big Five" are distributed to multiple services, compared to only 11.5% for smaller studios. This difference holds after controlling for title characteristics, estimated using the following regression:

$$E_j = \beta_0 + \beta_1 \cdot \text{BigFive}_j + \beta_2 \mathbf{X}_j + \varepsilon_j, \tag{6}$$

where E_j represents exclusive distribution, BigFive_j represents production by a "Big Five" studio, and $\mathbf{X}_j$ is a matrix of title characteristics including release years, type (movie or TV show), and genres. The results, presented in Column (1) of Table 1, confirm the distinction in the contracting strategies of the "Big Five" and small studios. This regression is used as an indirect inference moment when estimating the structural model.



(a) Services' Reliance on Third-Party Titles



(b) Licensee Choices: Walt Disney vs. Others

Figure 3: Descriptive Evidence on Title Distribution

Notes. Figure (a) displays the numbers of in-house and third-party titles on each streaming service within the sample. Figure (b) displays the likelihood of distributing to Hulu and its competitors, Netflix and Amazon Prime, by Walt Disney and all other studios. The vertical segments delimit the 95% confidence intervals.

Differential Reliance on Third-Party Titles by Streaming Services Streaming services differ in their reliance on third-party content. Figure 3a shows that third-party titles account for about two-thirds of the libraries of Netflix and Amazon, while Hulu depends almost entirely on licensing from external studios. In contrast, Disney Plus almost only offers in-house content from Disney-operated studios like Pixar and Marvel. This variation in reliance suggests that these services have different levels of vulnerability to the availability and exclusivity of third-party content, which can be affected by exclusive contracts.

Effect of Vertical Integration Studios often consider the benefits of their vertically integrated streaming services when licensing titles. As a result, a studio may prefer contracting with its integrated service over its competing services, as licensing to the latter can create negative externalities on its integrated service.[7] During the study period, Hulu—under Disney's control—was the only top service vertically integrated with non-in-house studios. Figure 3b shows that compared with other studios, Disney-affiliated studios are more likely to license titles to Hulu and less likely to license to Hulu's competitors, Netflix and Amazon Prime. This finding is confirmed by the following

[7] Vertically integrated streaming services must pay for titles from studios they control but do not directly operate. For example, Hulu pays for ABC's titles, though both being under The Walt Disney Company.

regression:

$$L_{jk} = \beta_0 + \beta_1 \cdot \text{WaltDisney}_j + \beta_2 \cdot \text{WaltDisney}_j \cdot \mathbf{1}(k = \text{Hulu}) + \delta_k + \varepsilon_{jk}, \tag{7}$$

where L_{jk} indicates if title j is licensed to service k, WaltDisney_j denotes Walt Disney-produced titles, and δ_k refers to service fixed effects. I present the results in Column (2) of Table 1, which are used as an indirect inference moment in structural estimation. A significantly positive β_2 reflects the impact of vertical integration on bargaining outcomes.[8]

5 Model

This section introduces a model of demand and supply to capture both forces of differentiation and bargaining behind exclusive contracts. On the supply side, in stage 1, streaming services bilaterally bargain with studios over title distribution and license fees, while setting subscription prices simultaneously. On the demand side, households subscribe to bundles of streaming services in stage 2, and household members choose titles to watch in stage 3. I present the model in reverse order of timing and discuss the key assumptions.

5.1 Demand

I develop the demand model to analyze the impact of title distribution changes on subscription demand. To achieve this, it incorporates a title viewership model to capture how title distribution affects viewership and a household subscription model to link expected viewership to subscription demand. Their details are as below.

5.1.1 Stage 3: Title Viewership

Each individual i can stream up to 50 hours per week. In each 30-minute period t during week w, she can watch a title j offered by her subscribed services. She derives the same utility from streaming a title, regardless of the service it is streamed on. Her utility from watching title j during period t is specified as

$$u_{ijt}^{T} = \mathbf{w}_{jw(t)} \boldsymbol{\beta}_i^{\mathrm{w}} + \beta_i^0 + \zeta_{jw(t)} + \epsilon_{ijt}, \tag{8}$$

[8] I rule out the possibility that services account for the benefits of their vertically integrated studios, which, if present, could also explain the observed distribution pattern. For a discussion of this assumption, see footnote 13.

where $\mathbf{w}_{jw}$ represents title-week characteristics, including genre, time since release, title fixed effects, seasonality (summer and holidays), and a binge-release dummy for shows defined as simultaneous releases of at least four episodes. The preferences for $\mathbf{w}_{jw}$, denoted as β^{w}, vary across individuals and differ between shows and movies. ζ_{jw} represents characteristics observed by consumers but not econometricians. ϵ_{ijt} is individual preference shocks assumed to be i.i.d. T1EV distributed. The utility of choosing the outside option is normalized as $u^T_{i0t} = \epsilon_{i0t}$. The random coefficients, $\boldsymbol{\beta}^{\mathrm{w}}_i$ and β^0_i, are specified as

$$\beta^0_i = \bar{\beta}^0 + \beta^0_d d_i + \beta^0_v v^0_i, \quad \text{and } \boldsymbol{\beta}^{\mathrm{w}}_i = \bar{\boldsymbol{\beta}}^{\mathrm{w}} + \boldsymbol{\beta}^{\mathrm{w}}_d d_i, \tag{9}$$

where d_i denotes demographic variables, including age (2–17, 18–44, and 45+ years old), gender (male and female), and race (white, African American, and others). v_i represents unobserved demographics that follow a standard normal distribution $N(0, 1)$.

With subscriptions to service bundle c, individual i can access the set of titles available on all services within the bundle, denoted as $\mathcal{J}_{cw} = \cup_{k\in c}\mathcal{J}_{kw}$. The share of her time spent on title j at week w can be specified as

$$s^T_{ijw|c} = \frac{\exp\left(\mathbf{w}_{jw}\boldsymbol{\beta}^{\mathrm{w}}_i + \beta^0_i + \zeta_{jw}\right)}{1 + \sum_{j'\in\mathcal{J}_{cw}} \exp\left(\mathbf{w}_{j'w}\boldsymbol{\beta}^{\mathrm{w}}_i + \beta^0_i + \zeta_{j'w}\right)}. \tag{10}$$

5.1.2 Stage 2: Streaming Service Subscriptions

At the beginning of each month, household h decides which bundle of streaming services to subscribe to. The utility derived by household h in market m, defined as a combination of DMA and month, from subscribing to bundle c is specified as:

$$u^S_{hcm} = V_{hcm}\alpha^V - p_{cm}(1 + \iota_h)\alpha^p_h + \mathbf{x}_c\boldsymbol{\alpha}^{\mathrm{x}} + \xi_{cm} + \varepsilon_{hcm}. \tag{11}$$

The first component, $V_{hcm}\alpha^V$, represents utility derived from content offered by services in bundle c. Here, content utility V_{hcm} is defined as

$$V_{hcm} = \sum_{i\in h} \kappa_i \sum_{t\in m} \mathbf{E}_\epsilon \left[\max_{j\in\{0\}\cup\mathcal{J}_{ct}} u^T_{ijt}\right], \tag{12}$$

which aggregates the expected utility derived by each household member from the outside option and viewing content on subscribed services, $\mathcal{J}_{ct}$, during all time periods t within the month. Since subscriptions are shared among household members, V_{hcm} accounts for all members' expected utilities, weighted by κ_i that reflects the decision-making power of i within the household. I assume κ_i varies across three age-gender groups: male adults, female adults, and children.

The second component, $p_{cm}(1+\iota_h)\alpha_h^p$, corresponds to the disutility from prices p_h and taxes $p_h\iota_h$ on service subscriptions. Tax rates ι_h vary by households based on their residential locations. The price coefficient is defined as $\alpha_h^p = \bar{\alpha}^p + \alpha_{inc}^p \cdot inc_h$, where inc_h denotes household income.

Additional elements in u_{hcm}^S include $\mathbf{x}_c$, which contains three bundle-specific dummies. The first two, for Amazon Prime and Hulu, account for add-on channel benefits, with the Amazon dummy also reflecting broader Prime benefits. The third dummy, for bundles that include both Hulu and Disney Plus, captures benefits from Walt Disney's Hulu-Disney Plus bundle promotions, such as free ESPN Plus access. ξ_{cm} represents demand shocks observed by households but not by econometricians. ε_{hcm} is i.i.d. T1EV distributed household preference shocks. The value of the outside option is normalized to $u_{h0m} = \varepsilon_{h0m}$.

The probability for household h in market m to subscribe to service bundle c is

$$s_{hcm}^S = \frac{\exp\left(V_{hcm}\alpha^V - p_{cm}(1+\iota_h)\alpha_h^p + \mathbf{x}_c\boldsymbol{\alpha}^{\mathrm{x}} + \xi_{cm}\right)}{1 + \sum_{g\in\mathbf{C}} \exp\left(V_{hgm}\alpha^V - p_{gm}(1+\iota_h)\alpha_g^p + \mathbf{x}_{gm}\boldsymbol{\alpha}^{\mathrm{x}} + \xi_{gm}\right)}, \tag{13}$$

where $\mathbf{C}$ is the set of all service bundles. Let $H_m(h)$ be the distribution of household preferences in market m. The market share of bundle c in market m is $s_{cm}^S = \int s_{hcm}^S dH_m(h)$. Combining with (10), title j's market share is $s_{jw}^T = \int \sum_c \mathbf{1}(j \in \mathcal{J}_{ct}) \cdot s_{h(i)cm(w)}^S \cdot s_{ijw|c}^T d\,(i)$, where $F(\cdot)$ is the nationwide distribution of preferences across individuals and their households.

Substitutability between Service Subscriptions The substitutability or complementarity between service subscriptions depends on the added utility of consuming them together, compared to each individually (Gentzkow 2007). A larger positive added utility implies stronger complementarity, while a larger negative value suggests stronger substitutability.

Three model components affect this added utility. First, decreasing marginal utility from accessing more content, driven by title substitutions, is captured by random coefficients in the viewership model (8). Second, bundle-specific dummy parameters in the subscription model (11) capture the additional benefits of bundle subscriptions, such as free ESPN Plus access with the Hulu and Disney Plus bundle. They reflect potential complementarity between services. Finally, unobserved demand shocks ξ_{cm} also affect the added utility.

5.2 Supply

In stage 1, studios and streaming services negotiate bilateral contracts, while streaming services set subscription prices simultaneously. I now discuss the equilibrium conditions for subscription price setting (given bilateral contracts) and bilateral contracting (given subscription prices), respectively.

5.2.1 Stage 1b: Subscription Price Setting

Over the 12-month study period, streaming service owners simultaneously set monthly subscription prices to maximize their payoffs, specified as

$$\Pi_K = \sum_{k \in O_k} \sum_{m} M_m \cdot (p_{km} + r_k) \cdot \mathbf{E}_{\zeta,\xi}[s^S_{km}(p_{km}, \mathcal{J}_k)] - \sum_{k \in O_k} \sum_{j \in \mathcal{J}_k} \tau_{jk}. \tag{14}$$

The first component accounts for the expected benefits from consumer subscriptions, and the second for lump-sum license fees, τ_{jk}, paid for all licensed titles $j \in \mathcal{J}_k$. O_k represents the three owners of the top four streaming services, including the Walt Disney Company that controls both Hulu and Disney Plus.[9] M_m is the number of households, and p_{km} is the subscription revenue per subscriber. r_k captures all per-subscriber profit beyond subscriptions, including: (1) marginal costs (e.g., data transfer costs and distribution fees paid to digital media players),[10] (2) advertising revenue, and (3) managerial heuristics to prioritize market shares and growth over short-term profitability.[11] The net direction of r_k is ambiguous: it could be negative due to marginal costs or positive due to advertising and growth incentives. r_k is assumed constant across all markets for each service.

I assume that streaming services consider the expected market shares $\mathbf{E}_{\zeta,\xi}[s^S_{km}(p_{km}, \mathcal{J}_k)]$, rather than the realized market shares, as commonly assumed in the literature. The rationale is that price changes are usually determined and pre-announced long before implementation. Consequently, at

[9] Though Comcast owned 33% stake in Hulu during the study period, it relinquished its control in Hulu to Disney effectively on May 14, 2019.

[10] Streaming services pay digital media players like Apple TV when users subscribe via these players.

[11] This prioritization reflects interests of investors and managers. Similar adjustments are used in studies of ride-hailing (Castillo 2022, Rosaia 2020) and online retail (Gutierrez 2021) to capture long-term platform incentives.

the time of pricing, streaming services do not know the exact values of the unobserved demand shocks, ζ and ξ, but only their distributions.

In practice, subscription prices are uniform nationwide and rarely adjusted: during the study period, Amazon Prime Video's price remained at \$8.99, while the other three services each raised prices once. To reflect this, I use the necessary conditions for optimal pricing: no streaming service can improve its payoff by changing its subscription prices uniformly across all months and markets, implying the following first-order conditions:

$$\sum_m M_m \cdot \mathbf{E}_{\zeta,\xi} s^S_{km} + \sum_m \sum_{k' \in O_k} M_m \cdot (p_{k'm} + r_{k'}) \frac{\partial \mathbf{E}_{\zeta,\xi} s^S_{k'm}}{\partial p_{km}} = 0. \tag{15}$$

5.2.2 Stage 1a: Bilateral Contracting

In the bilateral contracting stage, the production studio for each title j determines the intended distribution network, $\mathcal{K}_j$, the set of streaming services it plans to reach agreements with. The studio then negotiates bilaterally with these services over both inclusion in the network and *lump-sum* license fees, τ_{jk}. Building on the industry norms, I assume that first, license fees are negotiated on a per-title basis and paid as lump sums, as described in Section 2; and second, when licensing content, Walt Disney considers the payoffs of both Hulu and Disney Plus but allocates licensed content only to Hulu, as Figure 3a shows that Disney Plus rarely offers third-party content. I detail the extensive form bilateral contracting game in Appendix B.1.

Before describing how bargaining outcomes are determined, I specify the payoff considered by the studio of each title j as

$$\Pi_j = \underbrace{\sum\nolimits_{k \in \mathcal{K}_j} \tau_{jk}}_{\text{License Fees}} + \underbrace{\gamma \cdot \log\left(V_j(\mathcal{K}_j)\right)}_{\text{Logged Viewership}} + \underbrace{\sum\nolimits_{k \in \mathcal{K}_j} \nu_k(\mathcal{K}_j)}_{\text{Unobserved Preferences}} + \underbrace{\mu \cdot \sum\nolimits_{k \in O_j} \Pi_k}_{\text{Effects of VI}}. \tag{16}$$

In addition to license fees collected from streaming services, studios consider three factors. First, they value the title's expected viewership under network $\mathcal{K}_j$, $V_j(\mathcal{K}_j)$, as it generates buzz to attract investors for sequel productions.[12] I use the logarithms of viewership as this advantage diminishes for highly popular titles that are already well-funded.

[12]This is supported by anecdotal evidences, for example, Netflix considers its large and engaged subscriber base as an advantage compared to its competing services in contract negotiations with studios.

Second, studios may have preferences for specific services based on their long-term relationships. For example, Sony licenses its titles more often to Netflix due to their long-time cooperation. These preferences are captured by $\nu_{jk}(\mathcal{K}_j)$, which vary at the network-service level and are observed by studios but not econometricians. They are assumed to be i.i.d. and follow a normal distribution $N(0, \sigma_\nu^2)$.

Finally, studios factor in the payoffs of their vertically integrated services, O_j. This explains studios' tendency to license to these services, as shown in Figure 3b. The internalization parameter μ measures the extent to which studios incorporate these payoffs, with $\mu = 1$ implying complete internalization (Crawford et al. 2018).[13] However, intra-firm frictions, common in conglomerates like Walt Disney, may limit coordination between vertically integrated studios and streaming services, suggesting $\mu < 1$.[14] Therefore, I estimate μ to assess the extent of internalization.

License Fee Negotiations To determine negotiated license fees, I adopt the Nash-in-Nash with Threat of Replacement (NNTR) bargaining solution from Ho and Lee (2019), where studios can use streaming services excluded from bargaining as "threat of replacement" when bargaining with included services. However, I assume that streaming services cannot use this strategy.

Under the NNTR solution, the license fee τ_{jk} negotiated between the studio of title j and service k maximizes their Nash product under a constraint, conditional on the contracting outcomes of all other title-service pairs (including no contracts):

$$\tau_{jk} = \arg\max_{\tau} \underbrace{[\Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau, \boldsymbol{\tau}_{-jk}\})]}_{\text{Studio's Gain-from-Trade}}^{b_j} \underbrace{[\Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau, \boldsymbol{\tau}_{-jk}\})]}_{\text{Service's Gain-from-Trade}}^{1-b_j}, \tag{17}$$

$$\text{s.t.} \quad \underbrace{\Pi_j(\mathcal{K}_j, \{\tau_{jk}, \boldsymbol{\tau}_{-jk}\})}_{\text{Studio's payoff under } \mathcal{K}_j} \geq \max_{k' \notin \mathcal{K}_j} \underbrace{\left[\Pi_j\big((\mathcal{K}_j \setminus k) \cup k', \{\tau_{jk'}^{res}, \boldsymbol{\tau}_{-jk}\}\big)\right]}_{\text{Studio's payoff when replacing } k \text{ with } k'}. \tag{18}$$

Equation (17) represents the standard "Nash-in-Nash" bargaining solution, where $b_j \in [0, 1]$ is the bargaining parameter. A higher b_j indicates greater bargaining power for the studio to command

[13]Unlike Crawford et al. (2018), I assume that streaming services do not consider the benefits of their vertically integrated studios in (14). If they did, a service would be willing to accept a license fee larger than its incremental variable payoff for a contract with its integrated studio, when its the studio has strong bargaining power or outside options. This practice is hard to rationalize.

[14]Such frictions are documented across industries (Crawford et al. 2018, Cuesta, Noton and Vatter 2019, Chen, Yi and Yu 2023). Hortaçsu et al. (2024) also find that different arms of a conglomerate do not operate as a unitary decision-maker.

higher fees. I assume b_j varies only between the "Big Five" and small studios and remains constant for the same studio against all streaming services. In Appendix E.2, I also explore alternative bargaining parameter specifications with richer variations. The gains-from-trade for title j's studio and service k are

$$\Delta_{jk}\Pi_f(\mathcal{K}_j, \{\tau, \boldsymbol{\tau}_{-jk}\}) = \Pi_f(\mathcal{K}_j, \{\tau, \boldsymbol{\tau}_{-jk}\}) - \Pi_f(\mathcal{K}_j \setminus k, \boldsymbol{\tau}_{-jk}), f \in \{j, k\} \tag{19}$$

where $\mathcal{K}_j \setminus k$ represents bargaining breakdowns that lead to the exclusion of service k from title j's distribution network, and equivalent, exclusion of j from k's content library. $\boldsymbol{\tau}_{-jk}$ is the vector of license fees negotiated by other services for title j.

The constraint (18) deviates from the "Nash-in-Nash" solution and illustrates how exclusive distribution can improve a studio's bargaining leverage. By committing to distributing to a limited set of services, a studio can threaten an included service to license to an excluded service instead. Therefore, the negotiated fee paid by included service k, τ_{jk}, must ensure that the studio's profit is at least equivalent to the profit it would obtain from an excluded service k' at the latter's reservation fee. This reservation fee, $\tau_{jk'}^{res}$, is the minimum amount that k' would accept to replace k in $\mathcal{K}_j$ rather than remain excluded:

$$\Pi_{k'}\big((\mathcal{K}_j \setminus k) \cup k', \{\tau_{jk'}^{res}, \boldsymbol{\tau}_{-jk}\}\big) = \Pi_{k'}\left(\mathcal{K}_j \setminus k, \boldsymbol{\tau}_{-jk}\right). \tag{20}$$

Implications of the Bargaining Solution To illustrate the NNTR solution, I focus on the scenario where the studio of title j and service k are not vertically integrated. In Appendix B.3, I show that under the NNTR solution $\boldsymbol{\tau}^*$, the studio's gain-from-trade is

$$\Delta_{jk}\Pi_j(\mathcal{K}_j, \boldsymbol{\tau}^*) = \max\Bigg\{ \underbrace{b_j \cdot \Delta_{jk}\Pi_{jk}(\mathcal{K}_j)}_{\text{Nash Bargaining Outcome}}, \underbrace{\max_{k' \notin \mathcal{K}_j}\left[\Delta_{jk'}\Pi_{jk'}\left((\mathcal{K}_j \setminus k) \cup k'\right)\right]}_{\text{Outcome with Threat of Replacement}} \Bigg\}. \tag{21}$$

Here, $\Delta_{jk}\Pi_{jk}(\mathcal{K}_j) = \Delta_{jk}\Pi_j(\mathcal{K}_j, \cdot) + \Delta_{jk}\Pi_k(\mathcal{K}_j, \cdot)$ represents the bilateral surplus generated by the studio of title j and service k from reaching a licensing agreement, which they divide in the contract. This bilateral surplus is not a function of license fees because they are lump sum transfers and therefore cancel out.

Equation (21) reflects the studio's two bargaining strategies, each with distinct implications for distribution network formation that will be modeled later. With each contracting service, the studio can either secure a b_j share of the bilateral surplus through Nash bargaining, or extract an amount that equals its best outside option via the threat of replacement, where an excluded service would accept a deviating contract that surrenders all bilateral surplus to the studio. The studio chooses the strategy that leads to higher license fees. To improve its Nash bargaining outcome, a studio may opt for a network that maximizes the sum of bilateral surpluses with its contracting services. In contrast, to improve outside options, it may choose a narrower network, even though it may reduce the bilateral surplus. These trade-offs are also illustrated in the stylized model. Similar intuitions apply to vertically integrated firms, as discussed in Appendix B.3.

Equation (21) also highlights why the NNTR solution better fits the video streaming market than the standard "Nash-in-Nash" solution. The latter, considering only the first term on the right-hand side of (21), rules out the impact of excluded services on bargaining outcomes, conditional on bilateral surpluses. In contrast, the NNTR solution enables studios to leverage exclusive contracts to create licensing right scarcity, induce licensing competition among services, and negotiate higher fees, aligning more closely with observed practices.

Distribution Network Formation As the owner of a title, its studio has the discretion to select a set of streaming services, denoted as a distribution network $\mathcal{K}_j$, to reach agreements with. At equilibrium, a distribution network $\mathcal{K}_j$ must satisfy two necessary conditions: stability and optimality. I detail each in turn.

The stability condition requires that no streaming service within the network would benefit from unilaterally terminating its contract with the studio at the NNTR license fees $\boldsymbol{\tau}$: $\Delta_{jk}\Pi_k(\mathcal{K}_j, \boldsymbol{\tau}) \geq 0$. For non-vertically integrated studios, this is equivalent to:

$$\Delta_{jk}\Pi_{jk}\left(\mathcal{K}_j\right) \geq \Delta_{jk'}\Pi_{jk'}\left(\left(\mathcal{K}_j \setminus k\right) \cup k'\right), \forall j, k \in \mathcal{K}_j, k' \notin \mathcal{K}_j, \tag{22}$$

where $\Delta_{jk}\Pi_{jk}\left(\mathcal{K}_j\right)$ is the bilateral surplus between the studio and service k. This implies that an included service must generate a higher bilateral surplus than any excluded service. Otherwise, the included service would terminate the contract, as (21) indicates that it would incur a negative payoff to outbid a more efficient excluded service.[15] Implications of this condition for vertically integrated studios are discussed in Appendix B.4.

The optimality condition requires that no alternative, stable distribution network is more profitable for the studio of any title in equilibrium:

$$\Pi_j\left(\mathcal{K}_j, \boldsymbol{\tau}(\mathcal{K}_j, b_j)\right) \geq \Pi_j(\mathcal{K}'_j, \boldsymbol{\tau}(\mathcal{K}'_j, b_j)), \forall j, \forall \mathcal{K}'_j \text{ that is stable,} \tag{23}$$

where $\boldsymbol{\tau}(\mathcal{K}_j, b_j)$ denotes the vector of license fees under network $\mathcal{K}_j$ and bargaining parameter b_j. Here, b_j determines network optimality: as shown in Equation (21), studios with weaker bargaining power are more likely to use the threat of replacement strategy, thereby increasing their likelihood of exclusive distribution to improve their outside options.

5.3 Discussion of Modeling Assumptions

Static Demand I assume that subscription demand is static. However, consumers may be inert and fail to cancel subscriptions on time (Einav, Klopack and Mahoney 2023, Miller, Sahni and Strulov-Shlain 2023). Ignoring inertia could bias the estimation of the price coefficient α^p. However, the extent of inertia may be limited. Streaming services often cite the lack of demand stickiness as a profitability challenge (e.g., Wall Street Journal 2022, 2024). Moreover, as detailed in the next section, α^p is identified primarily from cross-market rather than cross-time variation, making it robust to potential consumer inertia.

Simultaneous Pricing and Contracting I assume that all contracts between studios and streaming services, as well as service subscription prices, are determined simultaneously. In practice, these decisions are made months before execution and are not often adjusted over time, even when the services acquire or lose major titles. Therefore, this timing assumption is more realistic than the alternative of sequential decision-making and is adopted by prior studies including Ho and Lee (2017) and Crawford et al. (2018).

Outside Option Values in Bargaining The equilibrium outcome of the bargaining model suggests that any excluded service is willing to surrender the entire bilateral surplus to a studio as its

[15] Appendix B.4 provides a detailed proof. This condition resembles Proposition 2 in Ho and Lee (2019).

outside option. If there exist fixed bargaining costs, the outside option value will be overestimated, as such costs may make complete concessions unprofitable for services or, in extreme cases, lead them to opt out of negotiations entirely. However, this concern is limited, as I focus on 1,145 most popular third-party titles, all with estimated annual license fees exceeding $1 million. Therefore, fixed bargaining costs, even if present, are unlikely to significantly affect outside options or the bargaining outcomes.

Static Bargaining I assume the studio-service bargaining is static, due to the lack of state dependence on both the supply and demand sides. On the supply side, studios face minimal switching costs between streaming services because titles are digitally distributed. On the demand side, even if subscription demand may be state dependent at a monthly level, this effect is much weaker at the yearly level, where bargaining takes place.

No Threat of Replacement by Streaming Services I assume that studios, but not streaming services, have the ability to use the threat of replacement in bargaining. In practice, studios make this threat credible through contractual commitments, including specifying exclusivity clauses or limiting the set of licensees. However, streaming services face challenges in making similar credible commitments to limit contractual partners, due to both the high dimensionality in their licensing choices, as well as unlimited "shelf space," which prevents them from exploiting resource scarcity to foster competition among studios and enhance bargaining leverage (Abreu and Manea 2024, Chambolle and Molina 2023).

6 Demand Estimation and Results

6.1 Estimation and Identification

I jointly estimate the demand for service subscriptions and titles due to their interdependence: viewers choose titles from content libraries of subscribed services, while households choose subscriptions based on the offered content. Joint estimation also addresses selection bias, as service subscribers are more likely to have stronger streaming preferences. For estimation, I use the generalized method of moments approach from Berry, Levinsohn and Pakes (1995) combined with the adapted nested fixed point algorithm from Lee (2013). Computational details are provided in Appendix D. Below, I discuss the data variations that identify key parameters and the moment conditions used to exploit them.

Identification of Title Demand The variance in streaming preferences related to observed demographics, governed by β_d^0 and $\boldsymbol{\beta}_d^{\mathrm{w}}$, is identified by the covariance between observed demographics and title viewership across genres and seasons. These covariances are observed from weekly viewership data segmented by age, gender, and race.

The substitution between titles identifies streaming preferences related to unobserved demographics, β_v^0. A large β_v^0 suggests titles are close substitutes, meaning adding a title to a library minimally increases total viewing time. In addition, the correlation between the average household viewership and the number of service subscribers also identifies β_v^0. A large β_v^0 implies that as services expand their libraries, later subscribers should exhibit lower streaming preferences and spend less time streaming than earlier ones.

Identification of Subscription Demand Variation in tax rates on streaming subscriptions across DMAs identifies the mean price coefficient $\bar{\alpha}^p$. As shown in the left panel of Figure 4, higher tax rates in DMAs lead to higher subscription costs, and therefore, lower subscription demand. In addition, the covariance between market shares and bundle sizes helps with its identification, as larger bundles imply higher subscription spending. The heterogeneity in the price coefficient, governed by α_{inc}^p, is identified by the correlation between household income and subscription demand across DMAs.

Cross-DMA covariance between household size and subscription demand identifies the content utility coefficient α^V. As shown in the right panel of Figure 4, DMAs with larger household sizes exhibit stronger subscription demand, since larger households derive greater content utility V_{hcm} from sharing subscriptions across household members. In addition, the subscription demand response to content library changes also identifies α^V, with a larger response implying a larger α^V.

The identification of household members' decision-making powers, κ, relies on two sources. First, the geographical covariance between demographic population shares and subscription demand, with a strong positive covariance indicating greater decision-making power. Second, the covariance between service subscriptions and demographic-specific viewership. For example, if kids' viewership

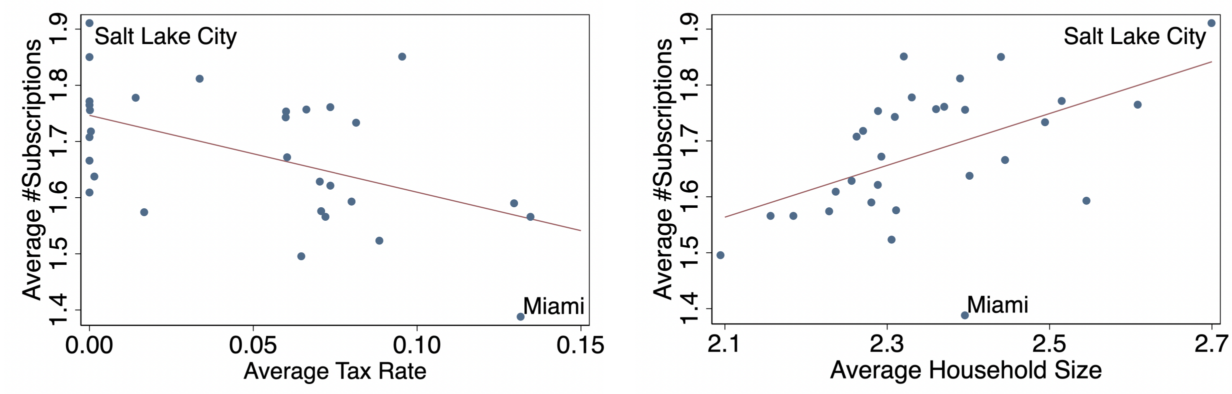


Figure 4: Demand for Streaming Service Subscriptions

Notes. This figure reports the correlation between the average numbers of subscribed top four services per household and two factors: tax rates and household sizes. Each dot corresponds to the average values of the variables in each of the top 30 DMAs during the study period. The red lines depict the first-order polynomial fits between the variables.

rises during summer without affecting subscription demand, it suggests limited decision-making power for kids.

Moment Conditions I adopt four sets of moment conditions. The first two sets apply to unobserved title and subscription demand shocks, ζ_{jw} and ξ_{cm}: $\mathbf{E}[\mathbf{Z}^T_{jw}\zeta_{jw}] = 0$ and $\mathbf{E}[\mathbf{Z}^S_{cm}\xi_{cm}] = 0$. Here, $\mathbf{Z}^T_{jw}$ and $\mathbf{Z}^S_{cm}$ are vectors of instrumental variables uncorrelated with ζ_{jw} and ξ_{cm}, respectively. I detail the selected instrumental variables below.

The remaining two sets are micro moments that match model-predicted and observed analogs (Petrin 2002). The third set matches demographic-specific viewership, conditional on title-week characteristics including genre and seasonality—for example, total viewing time spent by kids on dramas. These moments identify demographic-specific viewing preferences, governed by β^0_d and $\boldsymbol{\beta}^{\mathrm{w}}_d$. The last set matches covariances between subscription demand and gender-age group-specific viewership to identify decision-making parameters κ. When constructing the moments, subscription demand is measured by both the average number of subscriptions per household and the share of households without subscriptions.

Instrumental Variables In $\mathbf{Z}^T_{jw}$, I include the so-called "BLP instruments" to identify unobserved streaming preference heterogeneity, β^0_v. They are constructed by interacting the number of titles available on each service bundle with a dummy indicating the availability of each title on the bundle.

They reflect the level of competition for viewership faced by titles and are exogenous, under the assumption that the timing of title release and contracting precedes the realization of unobserved demand shocks ζ_{jw}.

To identify the mean price coefficient $\bar{\alpha}^p$, I include z^p in $\mathbf{Z}^S_{cm}$, defined as the interaction between bundle size and DMA-level mean tax rates.[16] This instrumental variable captures price variations from both bundle sizes and tax rates. It is exogenous to unobserved subscription demand shocks ξ_{cm} since streaming services set uniform national prices and generally do not adjust to local tax variations or bundle choices.[17] Tax rates are exogenous, as their applicability to streaming subscriptions is often decided by state courts and they apply broadly across industries, not specifically to video streaming.[18] I also interact z^p with average household income in each DMA to construct an additional instrument for identifying heterogeneity in price sensitivity (Miller and Weinberg 2017).

There are three more instrumental variables in $\mathbf{Z}^S_{cm}$. To identify the coefficient of content utility α^V, I include the average content portfolio value V_{hcm} of each bundle across all households in each market. To identify decision-making power parameters κ, I include the average shares of gender-age groups among the total population within each DMA.

6.2 Estimation Results

Title Demand I report the parameter estimates of demand for title viewership in Table 2. For computational feasibility, I set some coefficients of demographic-genre interactions, $\boldsymbol{\beta}^{\mathrm{w}}_d$, to zero. The estimates imply substantial heterogeneity in streaming preferences. On average, males, individuals over 45 years old, and non-white non-African Americans show less interest in streaming compared to their respective counterparts. Kids show strong preferences for kids and drama genres. In addition, the scale of β^0_v implies considerable preference heterogeneity in streaming that cannot be explained by observed demographics.

The estimates also imply that viewers' interest in a title decays over time since its release (Einav 2007), though this decay effect diminishes over time. In addition, viewers prefer streaming during

[16] Bundle size refers to the number of services within a bundle c in (11). DMA-level mean tax rates are weighted averages of ZIP code-level rates, using household numbers as weights.

[17] The only exception is the Hulu and Disney Plus bundle, which offers a bundled discount. However, the control variable for bundles including both Hulu and Disney Plus maintains the instrumental variable's validity.

[18] Special taxes like Florida's communication services tax and Chicago's amusement tax should also be exogenous, as they also apply to other businesses like broadband and event tickets.

Table 2: Demand Estimates: Titles

	Estimates	SE		Estimates	SE
Panel A: Heterogeneous Title Preferences					
Intercept: Standard Deviation	0.820	0.048	Genre: Thriller × Age: 45+	0.346	0.306
Intercept × Age: 2-17	−0.405	0.350	Genre: Thriller × Gender: Female	−0.073	0.019
Intercept × Age: 45+	−0.382	0.057	Genre: Kids × Age: 2-17	1.017	0.236
Intercept × Gender: Female	0.259	0.068	Genre: Kids × Gender: Female	0.058	0.054
Intercept × Race: Black	0.054	0.019	Genre: Drama × Age: 2-17	0.793	0.258
Intercept × Race: Others	−0.197	0.005	Genre: Drama × Age: 45+	−0.197	0.037
Genre: Action × Race: Black	0.045	0.237	Genre: Drama × Race: Black	−0.080	0.068
Genre: Comedy × Age: 45+	0.349	0.764	Season: Summer × Age: 2-17	0.206	0.160
Panel B: Other Title Demand Shifters					
Shows: Weeks Since Release	−0.116	0.006	Movies: Weeks Since Release	−0.190	0.020
Shows: Weeks Since Release^2	0.002	0.000	Movies: Weeks Since Release^2	0.002	0.000
Shows: Old (≥51 weeks)	−2.168	0.038	Movies: Old (≥51 weeks)	−3.937	0.117
Shows: Binge Release	0.573	0.144	Christmas and Thanksgiving	0.253	0.094

Notes. Title fixed effects are controlled. Reference group is Age: 18-44, Gender: Male, and Race: White.

Table 3: Demand Estimates: Service Subscriptions

	Estimates	SE
Price ($\bar{\alpha}^p$)	0.455	0.013
Price × HH income (α^p_{inc})	−0.061	0.005
Content Value (α^V)	0.846	0.286
Decision-Making Weights (κ): Female Adults	1.064	0.475
Decision-Making Weights (κ): Kids	0.160	0.180
Amazon	1.970	0.070
Hulu	0.655	0.093
Hulu Disney Plus Bundle	0.448	0.056
Intercept	0.817	0.117

Notes. Household income is in \$100,000 units and normalized to a mean of zero. κ for male adults is standardized to one.

holidays and binge-released shows.

Subscription Demand Table 3 reports parameter estimates of subscription demand. The estimates imply downward-sloping demand, with price sensitivity decreasing with household income. The estimated decision-making weights, κ_i, vary significantly across household members.[19] While female adults have similar decision-making weights to male adults, kids' weights are substantially lower and not significantly different from zero, implying their small impact on household subscription decisions.

I report own- and cross-price elasticities of subscription demand in Table 4. The average own-

[19]Because the average value of κ is not separately identifiable from α^V, I standardize κ to 1 for male adults.

Table 4: Demand Elasticities for Service Subscriptions

	Netflix	Amazon	Hulu	Disney
Netflix	-0.945	0.218	0.286	0.349
Amazon	0.117	-1.705	0.110	0.144
Hulu	0.111	0.080	-1.601	-0.178
Disney	0.107	0.083	-0.140	-1.568

Notes. The table presents elasticity of the demand for the column with respect to the price of the row.

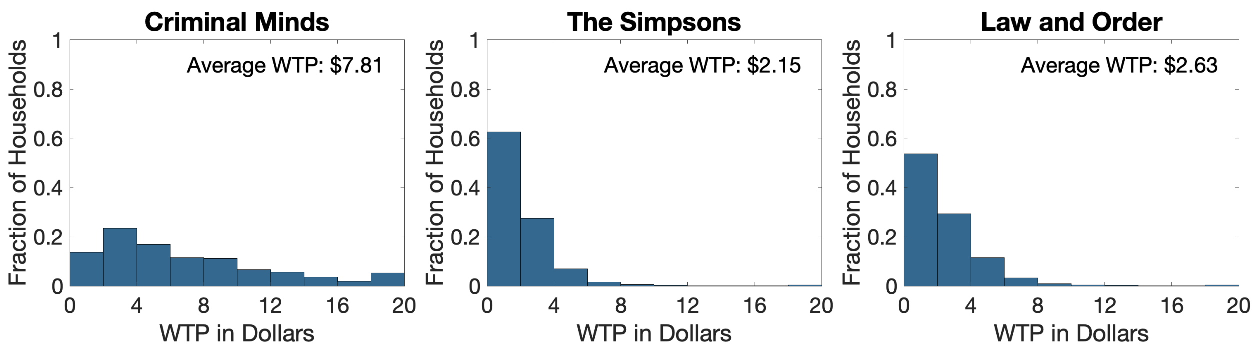


Figure 5: Household Willingness-to-Pay for Selected Titles

Notes. This figure displays the histograms of households' WTP for selected titles over March 2021 to February 2022, conditional on their access to all other titles. They have bins with a width of $2 and are right-censored at $20.

price elasticity is -1.45, while Netflix exhibits a notably lower elasticity of -0.94. This lower elasticity is partly attributed to its higher subscription prices, so its subscribers are less price-sensitive. Most pairs of streaming services are substitutes, since title substitution leads viewers to perceive diminishing marginal utility from additional subscriptions and accessing more content. However, Hulu and Disney Plus are complements, as Walt Disney's active promotion of their bundle creates demand synergy, increasing utility from subscribing to both (Gentzkow 2007).

Willingness-to-Pay (WTP) for Titles The key output of the demand model is households' WTP for each title, which determines how subscription demand responds to the addition or removal of this title from a service's content library.[20] Figure 5 presents histograms of the WTPs for three selected popular titles, showing significant variation across households for the same title. This variation is largely driven by demographics: larger, higher-income households tend to have higher WTPs due to greater content utility (V_{hcm}) and lower price sensitivity.

[20]To derive the WTP of each title, I evaluate its contribution to content utility V_{hcm} for each household, holding their access to other titles fixed. I then convert it to a dollar amount by multiplying with the estimated α^V/α_h^p.

The variance across titles is also significant: *Criminal Minds*, the most-viewed title in the sample, has an average WTP of \$7.81, while 57.2% (655) of sampled third-party titles have average WTPs below \$0.10. This variance results from differences in both quality and the audience groups they appeal to. For example, the average WTP for *Law and Order* (a crime thriller) is 22% higher than for *The Simpsons* (a kids' show), despite their similar expected viewership conditional on distributions. This discrepancy arises because most viewers of *The Simpsons* are kids, who, despite their strong preference, have limited decision-making power in their households.

7 Supply Estimation and Results

7.1 Estimation and Identification

On the supply side, I estimate three sets of parameters: (a) streaming services' per-subscriber profit beyond subscriptions, r_k, (b) bargaining parameters, $\boldsymbol{b}$, and (c) studio payoffs parameters, $(\gamma, \sigma_\nu, \mu)$. The estimation proceeds in two steps, with demand parameters taken as primitives. First, I estimate r_k using streaming services' first-order conditions (15). Second, I estimate $\boldsymbol{b}$ and $(\gamma, \sigma_\nu, \mu)$ using simulated methods of moments, whose identification relies on the variation in distribution networks of titles. I discuss the identification and estimation of $\boldsymbol{b}$ and $(\gamma, \sigma_\nu, \mu)$ below, with computational details in Appendix E.1.

Identification The difference between a studio's observed distribution networks and those that maximize the total bilateral surplus with its contracting services identifies the studio's bargaining parameter b, with a larger difference indicating a smaller b. This is because studios with strong bargaining power choose networks that maximize the bilateral surplus in negotiations, as they can extract most of this surplus. Conversely, studios with weaker bargaining power may narrow their networks and opt for exclusive distribution to improve their outside options.

Studio payoff parameters are identified as follows. Studios' preference for title viewership, γ, is identified by their tendency to license to Netflix, which has the largest and most engaged subscriber base, over other services. The internalization parameter for vertically integrated studios, μ, is identified by the tendency of Disney-affiliated studios to license to Hulu while foreclosing its competitors. The standard deviation of unobserved preferences, σ_ν, is primarily identified by the stability condition (22). A low σ_ν implies a strong correlation between a service's inclusion in a title's network, with its ability to generate high incremental variable payoffs relative to competitors through licensing the title.

Moment Conditions I use two sets of moment conditions. The first set is instrumental variable moments applied on the discrepancy between simulated likelihoods and observed choices of distribution networks for each title (Pakes and Pollard 1989):

$$\mathbf{E}[(\hat{P}_{j\mathcal{K}} - D_{j\mathcal{K}})\mathbf{Z}_{j\mathcal{K}}] = 0, \forall j, \mathcal{K}. \tag{24}$$

Here, $\hat{P}_{j\mathcal{K}}$ and $D_{j\mathcal{K}}$ are the simulated probability and observed binary indicator, respectively, for title j to be distributed to network $\mathcal{K}$. $\mathbf{Z}_{j\mathcal{K}}$ is a vector of instrumental variables. It includes network-specific dummies and logged title viewership under different networks, which identify $\boldsymbol{b}$ and γ. To identify σ_ν, $\mathbf{Z}_{j\mathcal{K}}$ also includes differences in the incremental variable payoffs from licensing each title j across services under each network $\mathcal{K}_j$:

$$\sum_{k\in\mathcal{K}} \sum_{k'\notin\mathcal{K}} \left(\mathbf{E}_\nu \left[\Delta_{jk}\Pi_k(\mathcal{K}_j)\right] - \mathbf{E}_\nu \left[\Delta_{jk}\Pi_{jk'}\left((\mathcal{K}_j \backslash k) \cup k'\right)\right]\right). \tag{25}$$

The second set includes indirect inference moments that match regression results using simulated and observed data (Gourieroux, Monfort and Renault 1993). I consider two linear probability regressions: (a) whether a title is exclusively distributed, based on production by the "Big Five" or smaller studios, to identify the disparity in $\boldsymbol{b}$ between these groups; and (b) the probability of licensing to Hulu versus Netflix and Amazon Prime for Disney-affiliated versus other studios, to identify the internalization parameter μ. Regression results using observed data are in Table 1.

7.2 Estimation Results

In Table 5, I present the estimated supply parameters. Studios value high title viewership, as doubling a title's viewership is valued at 0.77 million by its production studio. This emphasis on viewership offers Netflix a competitive edge in licensing titles due to its larger and more engaged subscriber base. The estimated internalization parameter μ suggests that for every dollar earned by a streaming service, its vertically integrated studios perceive only \$0.63. This incomplete internalization may be attributed to internal frictions between the content production and streaming service divisions within conglomerates.

Table 5: Estimation Results: Supply

	Estimates	SE
Bargaining Parameters *b*		
"Big Five"	0.819	0.035
Small studios	0.534	0.192
Studio Payoff Parameters		
Viewership preference γ	0.775	0.184
STD of unobserved preferences σ_ν	0.147	0.025
Internalization μ	0.627	0.137

Notes. Studios' payoffs are measured in millions of dollars. Standard errors are computed using 100 bootstrap samples.

The estimation results predict that studios, on average, receive an annual license fee of \$9.22 million per title, representing 90.3% of the incremental variable payoff generated by streaming services from licensing the title. These significant fees are driven by studios' large bargaining parameters (above 0.5) and their ability to use excluded streaming services as bargaining leverage. This large magnitude of license fees also suggests that the variance in studios' unobserved contracting preferences (σ_ν) is not economically significant.

Furthermore, the estimation highlights that the "Big Five" studios have significantly stronger bargaining power than small studios. This stronger power of the "Big Five" likely comes from their long histories and extensive content libraries, which give them both negotiation expertise to better assess their content's value and set appropriate license fees, as well as leverage that leads streaming services to make concessions to preserve long-term relationships and avoid being foreclosed in future licensing deals.

Robustness Checks In Appendix E.2, I conduct robustness checks as follows. To verify the profit margin of Amazon Prime Video subscriptions recovered using its nominal price of \$8.99 and the first-order conditions (15), I re-estimate the margin together with $(\boldsymbol{b}, \gamma, \sigma_\nu, \mu)$ and identify it through title distribution variation. To ensure that the main bargaining parameter specification captures most of its variation, I examine alternative specifications, including different studio categorizations and varying bargaining parameters across streaming services. Finally, to rule out the impact of any potential costs for titles to switch across services, I re-estimate the supply model using only newly released titles. Results from all robustness checks closely align with those in Table 5, supporting the model's validity.

Model Fit To assess out-of-sample fit, I compare the estimated and reported annual license fees for Netflix and Hulu. The model predicts Netflix's fees at \$5.9 billion and Hulu's at \$3.5 billion, which closely align with their reported expenses of \$5.0 billion and \$3.3 billion, respectively, even though these reported numbers are not used in estimation.[21] In Appendix E.3, I also examine in-sample model fit, which shows a close alignment between the model's predictions and actual data on subscription prices, consumer demand, and title distributions. Together, these close alignments suggest good model fit.

8 Counterfactuals: The Impact of Exclusive Contracts

In this section, I apply the estimates to evaluate the impact of exclusive contracts on studios, streaming services, and consumers. First, I investigate the short-run impact by simulating a counterfactual without exclusive contracts, keeping the sets of competing services and produced titles fixed as observed. Next, I explore their potential long-term effects if they reshape competition in streaming service and content production markets.

8.1 The Short-Term Impact of Exclusive Contracts

Counterfactual Design I examine a counterfactual scenario without exclusive contracts, where the current array of streaming services and titles remains unchanged. In this scenario, studios retain the ability to choose distribution networks but cannot commit to any distribution arrangements through exclusivity clauses. Without such commitments, studios can always negotiate additional agreements with previously excluded services. Beyond this design, I explore an alternative in Appendix F, where all third-party titles must be distributed across all streaming services.

To reflect studios' loss of ability to make contractual commitments, I apply the following equilibrium refinement condition, that under an equilibrium distribution network $\mathcal{K}_j$, no mutually beneficial

[21] Netflix only reports global license fees. I approximate its U.S. fees by multiplying the share of U.S. revenue with its global fees. While Hulu operates only in the U.S., it last disclosed its content spending in 2020, which is used for comparison. Amazon Prime only reports its license fees for music and video streaming businesses together, so a comparison is not possible.

bilateral contracts exist between the studio and any set of services outside of $\mathcal{K}_j$.[22] This condition rules out any profitable deviations for studios and excluded streaming services. Specifically, for any set of excluded services K'_j with $K'_j \cap \mathcal{K}_j = \varnothing$, and any bilateral contracts between the studio and K'_j with license fees τ', at least one of the following must hold: the studio does not benefit from deviating to accept these contracts:

$$\Pi_j(\mathcal{K}_j, \tau) \geq \Pi_j(\mathcal{F}(\mathcal{K}_j \cup K'_j), \{\tau, \tau'\}), \tag{26}$$

or at least one streaming service in K'_j does not benefit:

$$\exists k' \in K'_j, \text{s.t. } \Pi_{k'}(\mathcal{K}_j, \tau) \geq \Pi_{k'}(\mathcal{F}(\mathcal{K}_j \cup K'_j), \{\tau, \tau'\}). \tag{27}$$

Here, the function $\mathcal{F}$ adjusts the distribution network after the studio's deviation to capture a service's ability to unilaterally terminate its existing contract, a refinement condition used in the main model, as well as Ho and Lee (2019) and Ghili (2022). A service will decide to terminate its contract if its profitability falls below zero. Any service opting for termination will not be part of the adjusted network, $\mathcal{F}(\mathcal{K}_j \cup K'_j)$. Studios can still leverage excluded streaming services as a bargaining threat against included services, provided the distribution network is stable under the new refinement condition.

This refinement condition highlights the trade-off in exclusive contracts: while studios lose the ability to contractually commit to networks that maximize total bilateral surpluses with contracting services, reducing contracting efficiency (Bernheim and Whinston 1998, Segal 1999), the reduced stability of exclusive networks helps mitigate inefficient exclusions, which arise due to studios seeking to improve their outside options for bargaining.

Simulating counterfactual outcomes requires finding subscription prices, title distribution networks and the implied license fees, and consumer demand that are optimal against each other. The interdependent nature of these decisions creates an extensive action space, making direct enumeration of all equilibria computationally infeasible. To address this, I apply the algorithm from

[22] This refinement condition is weaker than the "pairwise stability" condition in Jackson and Wolinsky (1996) and Ghili (2022), which only allows a studio to deviate to contract with one excluded streaming service. This condition is also better aligned with the supply model, where a studio can reach agreement with any set of services.

Lee and Pakes (2009), where studios sequentially select the payoff-maximizing distribution network for each third-party title, conditional on others' distributions. Simultaneously, streaming services adjust subscription prices in response to updated networks. This procedure repeats until there are no further adjustments in networks or subscription prices, reaching a simultaneous-move Bayesian Nash equilibrium for a given draw of unobserved contracting preferences ν. I perform this algorithm for 30 random draws of ν and average results across them.

I present the outcomes in Column (1) of Table 6. Comparing the counterfactual scenario to the status quo, exclusive contracts increase the share of titles distributed to only one service by 26.6 percentage points, indicating many currently exclusive distribution networks rely on contractual commitment for sustainability. The results suggest that the impact of exclusive contracts varies significantly among services and studios.

Impact on Streaming Services Small streaming services like Hulu, which typically lack in-house content, benefit significantly from exclusive contracts. In particular, Hulu's payoff substantially increases by 110.1%. This is due to their reliance on exclusive third-party content to differentiate from competitors, attract subscribers, and enhance pricing power. In addition, this increase in profitability strengthens small services' positions in licensing negotiations, as they can become more profitable contracting partners to "outbid" streaming giants for exclusive rights. This increased competitiveness in licensing unique content further strengthens small services' appeal to consumers.

In contrast, larger streaming services see minimal gains or even losses. Netflix gains a modest 1.6%, while Amazon Prime sees a 5.3% loss. This is because they already have substantial unique, in-house content that differentiates them from competitors. As a result, their incremental benefit from further differentiation through exclusive third-party content is limited. Instead, exclusive contracts lead to negative equilibrium effects for these streaming giants, as small services like Hulu use exclusive third-party content to attract subscribers away from them. This increased competition offsets their modest gains from increased differentiation, resulting in Netflix's minimal gain and Amazon Prime's loss.

Impact on Studios The "Big Five" studios see a 5.7% decrease in their payoffs. This is due to a negative equilibrium effect, or "contracting externalities:" when a streaming service secures an

Table 6: Short-Term Impact of Exclusive Contracts

		(1) Simulated Status Quo	(2) No Excl. Contracts	%Change from (2) to (1)
Distribution Networks	Netflix	0.755	0.866	−12.9%
(%Third-Party Titles on)	Amazon Prime	0.229	0.441	−47.9%
	Hulu	0.222	0.440	−49.6%
	Only One Service	0.829	0.563	47.4%
Subscription Prices	Netflix	14.678	15.854	−7.4%
(Average $ Per Month)	Amazon Prime	9.546	9.420	1.3%
	Hulu	8.204	6.174	32.9%
	Disney Plus	7.829	7.868	−0.5%
Market Shares	Netflix	0.556	0.556	0.0%
(Average Per Month)	Amazon Prime	0.443	0.450	−1.7%
	Hulu	0.289	0.209	38.2%
	Disney Plus	0.299	0.302	−1.1%
	Multi-Homing	0.414	0.364	13.6%
Service Payoffs	Netflix	5.940	5.844	1.6%
($Bn Per Year)	Amazon Prime	2.416	2.552	−5.3%
	Hulu	0.331	0.158	110.1%
	Disney Plus	2.056	2.095	−1.9%
	Total Service Payoff	10.744	10.649	0.9%
Studio Payoffs	Big Five	8.470	8.981	−5.7%
($Bn Per Year)	Remaining	4.361	4.036	8.1%
	Total Studio Payoff	12.831	13.017	−1.4%
Consumer Surplus ($ Per HH-Year)		204.476	228.682	−10.6%
Avg. Weekly Streaming Hours		2.537	2.715	−6.6%

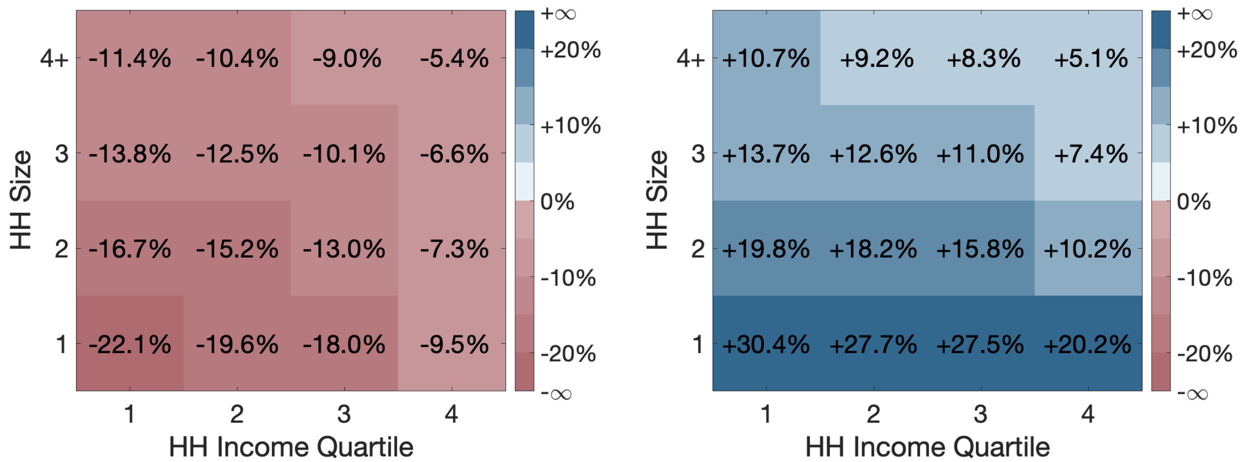
Notes. Specification (1) simulates the status quo, while (2) removes exclusive contracts while keeping titles and streaming services unchanged. The last column reports percentage changes from specification (2) to (1). The share of multi-homing households includes those with at least two subscriptions to Netflix, Amazon Prime, or Hulu (excluding Disney Plus).

exclusive contract for one title, its willingness-to-pay for other titles diminishes. This is because unique content increases differentiation and reduces substitutability among services, making their demand less responsive to changes in content libraries. Consequently, a streaming service sees smaller subscriber losses after losing a title. This effect reduces the service's willingness-to-pay for the title, and consequently, the bilateral surplus associated with its contract. Given their strong bargaining power, the "Big Five" studios can extract most of these bilateral surpluses in bargaining, whether or not exclusive contracts can be used as a bargaining tool. As a result, the overall decrease in bilateral surpluses lowers the license fees that the "Big Five" can demand, reducing their profitability.

However, small studios see an 8.1% increase in payoffs, despite facing the same negative equilibrium effect as the "Big Five." This differential impact is due to their lower bargaining power (0.53 compared to 0.82 for the "Big Five"). Exclusive contracts facilitate exclusive distributions, allowing studios to use excluded streaming services as bargaining threats. In addition, the improved profitability of small services makes them credible threats in negotiations against Netflix and Amazon Prime. Both factors enhance small studios' bargaining leverage, offsetting the negative equilibrium effect and resulting in higher license fees.

Impact on Consumers Exclusive contracts reduce consumer surplus by $24.2 per household per year, equivalent to a 10.6% decline. This loss results from two sources. First, reduced title distribution forces households to subscribe to more services to access their content of interest—the share of households that multi-home across Netflix, Amazon Prime, and Hulu increases by 5.0 percentage points under exclusive contracts. Second, softened competition leads to higher subscription prices. Even holding title distribution fixed at the status quo, price increases alone reduce consumer surplus by $10.8 per household per year.

The negative impact of exclusive contracts, when measured as a percentage of household surplus, is highly regressive. As Figure 6a shows, smaller and poorer households face disproportionately large losses. Larger households, despite valuing content more, lose less because they are more likely to subscribe to multiple services concurrently, so they can access most titles regardless of whether their distributions are exclusive or not. Poorer households lose more due to greater price sensitivity, making them vulnerable to subscription price increases, especially by Hulu, whose low prices appeal



(a) Short-Term

(b) Long-Term: Increased Service Entry

Figure 6: The Distributional Impact of Exclusive Contracts on Consumers

Notes. This figure shows the distributional impact of exclusive contracts on consumers (households) measured in percentage points, comparing the status quo with a scenario without these contracts. All counterfactuals remove exclusive contracts while: (a) keeping titles and streaming services the same as in the status quo, abd (b) assuming Hulu's exit. Households are categorized by sizes (1, 2, 3, and 4 or more) and income quartiles (1: less than $52,000; 2: $52,000–$89,599; 3: $89,600–$150,259; 4: $150,260 and above).

to poorer households. In Figure A.3, I also describe the distributional effect in dollar amounts. When measured in absolute terms, larger and wealthier households generally lose more because they value content more, and therefore, lose more from reduced title distribution.

8.2 The Potential Long-Term Impact of Exclusive Contracts

The analyses above show that exclusive contracts hurt consumers when the set of streaming services and produced titles is held constant. However, they also improve the profitability of small studios and small streaming services, suggesting that over the long run, exclusive contracts can potentially encourage content production and the entry of streaming services. To explore these possibilities, I analyze two additional counterfactual scenarios and discuss their implications below.

8.2.1 Increased Title Production

I begin by quantifying the impact of exclusive contracts when they stimulate content production by small studios. To do this, I simulate counterfactuals without exclusive contracts, assuming that only a fraction of the titles produced by small studios under the status quo would have been created otherwise. In each of 150 simulations, I randomly (a) draw a production share uniformly from $(0, 1)$,

Table 7: Consumer Surplus Without Exclusive Contracts: Impact of Small Studio Production

	(1) No Price Control		(2) With Price Control	
	Estimate	SE	Estimate	SE
Share of Small Studio Titles Produced	37.034	4.496	66.811	0.961
Mean Subscription Prices			−40.019	0.379
Intercept	192.818	7.517	555.196	3.474
Observations	150		150	

Notes. Each observation corresponds to a counterfactual simulation. The dependent variable is consumer surplus, measured in dollars per household per year.

(b) select which small studios' titles are produced according to that share, and (c) draw studios' unobserved contracting preferences ν. As a caveat, the simulated consumer surpluses should be interpreted as lower bounds: in practice, studios likely have some prior knowledge of title quality at the time of production, so the smaller set of titles produced under the counterfactuals would likely have higher average quality than assumed in the simulations.

To investigate the impact of increased title production on consumer surplus, I regress simulated consumer surplus on small studios' production shares. Column (1) of Table 7 presents the regression results, and Figure 7 describes the fitted values. I find that, on average, a 10% increase in small studio production—equivalent to 61 additional titles—is associated with a \$3.7 gain in consumer surplus per household per year. This estimate implies that consumers are better off without exclusive contracts than under the status quo as long as small studios retain at least 31% of their current production. Since studios likely have some prior knowledge of title quality and would prioritize higher-quality titles when production is constrained, this threshold is likely overestimated. Therefore, exclusive contracts would need to at least triple small studio production to be welfare enhancing for consumers, a scale of effect that is unlikely to realize in practice.

Why, then, is the consumer gain from increased content production so limited? To explore this, I run a second regression that adds average simulated subscription prices across all service-months as a control. Column (2) of Table 7 reports the results. After controlling for price, a 10% production increase is associated with a \$6.7 rise in consumer surplus—nearly double the estimate without the control. This comparison implies that for consumers, roughly 45% of the potential gains from increased content are offset by higher subscription prices, as streaming services use new content to differentiate and raise prices.

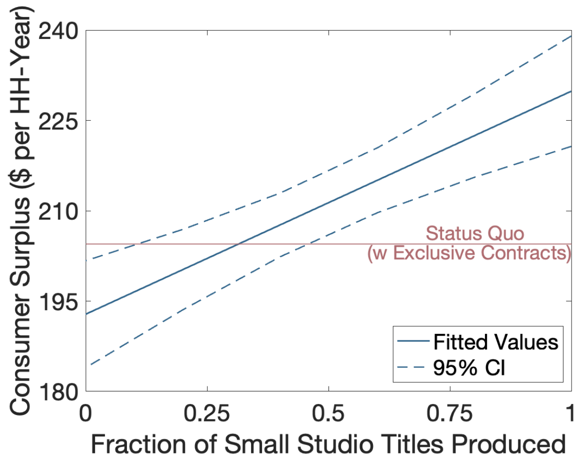


Figure 7: Consumer Surplus without Exclusive Contracts and with Varying Small Studio Content Production

Table 8: Long-Term Impact of Exclusive Contracts: Increased Service Entry

	(1) Simulated Status Quo	(2) No EC & No Hulu Entry	%Change from (2) to (1)
Share of Third-Party Titles on Only One Service	0.829	0.890	−6.9%
Total Service Payoff ($Bn Per Year)	10.744	11.210	−4.2%
Total Studio Payoff ($Bn Per Year)	12.831	12.495	2.7%
Consumer Surplus ($ Per HH-Year)	204.476	183.302	11.6%
Avg. Weekly Streaming Hours	2.537	2.482	2.2%

Notes. Specification (1) simulates the status quo, while (2) removes exclusive contracts and Hulu. The last column reports percentage changes from specification (2) to (1).

8.2.2 Increased Service Entry

The large positive impact of exclusive contracts on small services like Hulu suggests they may benefit consumers by enabling the entry of small services that might otherwise be unprofitable to enter and operate. To assess this potential, I quantify the impact of exclusive contracts when they stimulate the entry of small services. Table 8 presents key counterfactual outcomes in which exclusive contracts are removed alongside Hulu, with full results in Table A.3.

When exclusive contracts encourage the entry of small services like Hulu, they can lead to less-exclusive distribution networks. The share of exclusively distributed titles falls by 6.1 percentage points, as studios must exclude more services to maintain exclusivity, foregoing additional revenue and thereby weakening the incentive to exclude.

The entry of small services intensifies competition among streaming services for consumers, driving down subscription prices and lowering the total service payoff by 4.2%. At the same time, this heightened competition raises platforms' willingness-to-pay for content licensing to attract and retain subscribers, resulting in a 2.7% increase in studios' payoffs.

Less restrictive title distribution and lower subscription prices increase consumer surplus by $21.2 per household per year, an 11.6% gain.[23] Figure 6b shows that lower-income, smaller households—those most harmed by exclusive contracts under a fixed set of streaming services—benefit the most from the entry of small services, since these entrants provide smaller, affordable content libraries tailored to the low streaming needs and high price sensitivity of these consumers.

This counterfactual exercise underscores that exclusive contracts can generate significant and progressive long-run benefits by intensifying competition among streaming services. Compared to the limited gains from increased title production discussed in the previous subsection, these results suggest that downstream competition has a more substantial impact on consumers than upstream competition. This finding aligns with insights from Rey and Tirole (2007, Section 2.1.4): the social benefits of increased competition among intermediaries (streaming services) are more likely to pass through to consumers, while increased upstream (studio) competition tends to be absorbed by intermediaries, limiting the gains perceived by consumers.

9 Conclusions

In this paper, I study the welfare implications of exclusive contracts in the video streaming market. I develop a model that combines demand for titles and subscriptions, subscription price setting by streaming services, and contract negotiations between studios and services. In particular, I identify studios' bargaining power without data on negotiated licensing fees, instead using the relationship that studios with weaker bargaining power are more inclined to opt for exclusive distribution to improve their bargaining leverage.

To quantify the effect of exclusive contracts, I apply estimates to simulate a counterfactual without these contracts. I find that the effects vary significantly among services and studios. Small streaming services, such as Hulu, benefit substantially due to reliance on exclusive third-party titles for differentiation, whereas Netflix and Amazon Prime see minimal or negative impacts. Though the "Big Five" studios lose from exclusive contracts, small studios gain due to the increased bargaining

[23] Hulu's entry and exit should have minimal impact on the value of possibly available content for consumers, as it has only 16 in-house titles, none with an average household WTP exceeding $0.5 per year. All of Hulu's third-party titles in the status quo are licensed to Netflix or Amazon Prime Video in the counterfactual.

leverage. Consumers lose in the short run, but this loss may get mitigated or reversed in the long run due to stimulated content production and streaming service entries.

This paper can be extended in several directions. One direction is the explicit modeling of title production to understand how exclusive contracts affect content production, which is not directly addressed in this paper due to studios' capacity constraints during the study period. In addition, the video streaming market relies on sectors like cloud computing for data storage, ISPs for data transmission, and digital media players for content distribution. Vertical integration is common in these sectors, such as Amazon's ownership of Amazon Cloud in cloud computing and Fire TV in digital media players. Research into these vertical relationships and their interaction with studio-streaming service dynamics could shed light on the broader implications of vertical integration.

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Appendix A Additional Figures and Tables

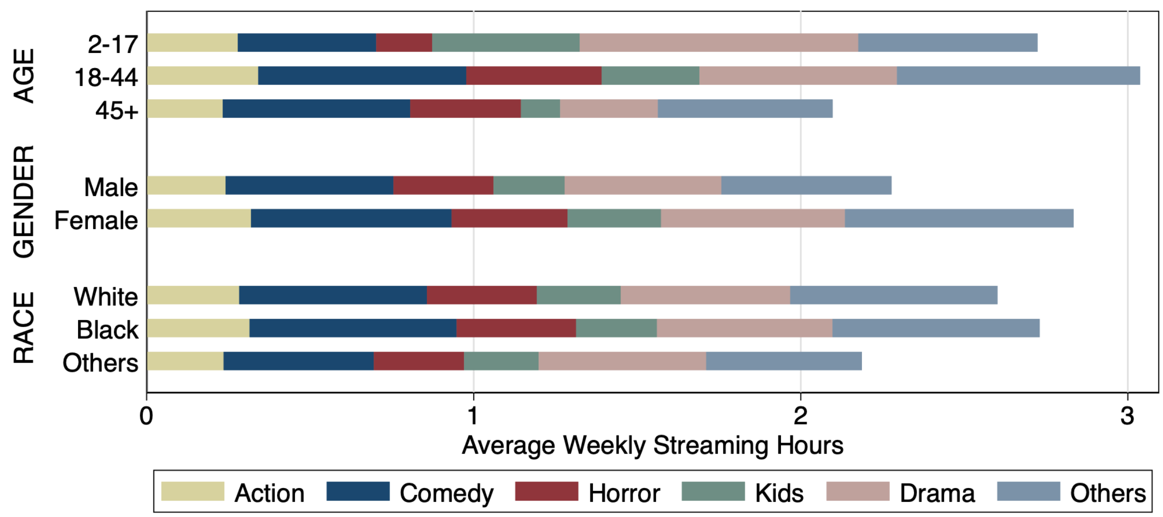


Figure A.1: Average Weekly Streaming Hours by All Genres and Demographic Groups

Notes. This figure displays the average weekly streaming hours across six different genres for U.S. individuals, categorized by their age, gender, and race.

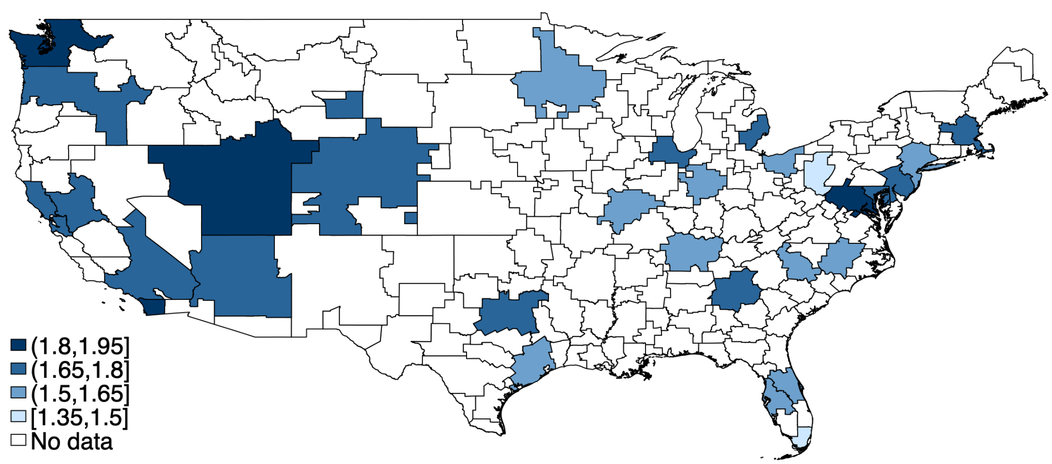
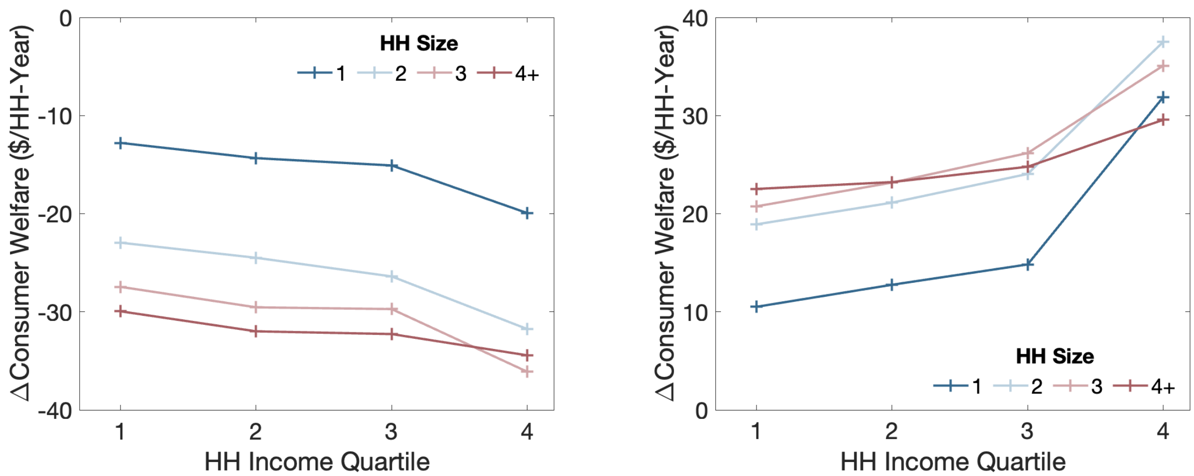


Figure A.2: Subscription Demand in Top 30 DMAs

Notes. This This map shows the average number of top four streaming services subscribed to by households for the 30 most populous DMAs from March 2021 to February 2022.



(a) Short-Term

(b) Long-Term: Increased Service Entry

Figure A.3: The Distributional Impact of Exclusive Contracts on Consumers in Dollar Amounts

Notes. This figure displays the percentage points change in household surplus across income levels and household sizes from each of the three counterfactual scenarios to the status quo. All counterfactuals remove exclusive contracts while: (a) keeping titles and streaming services the same as in the status quo, (b) assuming a one-third reduction in content production by small studios, and (c) assuming Hulu's exit. Households are grouped by sizes (1, 2, 3, and 4 or more) and income quartiles (1: less than $52,000; 2: $52,000–$89,599; 3: $89,600–$150,259; 4: $150,260 and above).

Table A.1: Descriptive Evidence: Demand for Streaming Service Subscriptions

	#Subscribed Services		Spending: Pre-Tax	
	(1)	(2)	(1)	(2)
Tax rate	-1.417	-1.069	-13.067	-9.486
	(0.447)	(0.522)	(3.934)	(4.526)
Household size		0.345		3.552
		(0.124)		(1.101)
Constant	1.749	0.923	17.782	9.258
	(0.026)	(0.304)	(0.240)	(2.703)
#Observations	360	360	360	360

Notes. Observations are at DMA-month level. Standard errors are clustered at DMA level and reported in the parentheses. The two dependent variables are the average number of subscriptions and the total pre-tax subscription spending per household.

Table A.2: Summary Statistics: Titles

	N. Obs.	Mean	Std. Dev.	Median	5th Pctile.	95th Pctile.
Among All Titles						
Viewership (million hours)	2,028	20.356	40.569	8.086	1.642	77.136
Genre: Action	2,028	0.169	0.375	0.000	0.000	1.000
Genre: Comedy	2,028	0.176	0.381	0.000	0.000	1.000
Genre: Horror/Thriller	2,028	0.148	0.356	0.000	0.000	1.000
Genre: Kids	2,028	0.078	0.269	0.000	0.000	1.000
Genre: Drama	2,028	0.223	0.417	0.000	0.000	1.000
Genre: Others	2,028	0.206	0.404	0.000	0.000	1.000
Among Third-Party Titles						
Viewership (million hours)	1,145	21.684	45.238	7.970	1.863	83.580
"Big Five"	1,145	0.470	0.499	0.000	0.000	1.000
Available on One Service	1,145	0.867	0.339	1.000	0.000	1.000
Available on Two Services	1,145	0.109	0.312	0.000	0.000	1.000
Title-Week Level Statistics						
Viewership (million hours)	82,693	0.499	1.426	0.162	0.009	1.875
Viewership: Netflix Titles	54,267	0.496	1.506	0.151	0.008	1.867
Viewership: Amazon Titles	12,024	0.503	1.253	0.156	0.008	2.133
Viewership: Hulu Titles	14,146	0.579	1.365	0.202	0.014	2.310
Viewership: Disney Plus Titles	9,883	0.583	1.417	0.263	0.022	1.904
TV Shows: Binge Release	48,504	0.428	0.495	0.000	0.000	1.000

Notes. The table reports summary statistics for the period between March 2021 and February 2022. It covers 2,028 titles, including 1,145 third-party titles. 88 titles are missing production data to determine if they are third-party titles. Binge release is defined as simultaneous releases of at least four episodes.

Table A.3: Long-Term Impact of Exclusive Contracts: Increased Service Entry (Complete Results)

		(1) Simulated Status Quo	(2) No EC & No Hulu Entry	%Change from (2) to (1)
Distribution Networks	Netflix	0.755	0.811	−7.0%
(%Third-Party Titles on)	Amazon Prime	0.229	0.299	−23.2%
	Hulu	0.222		
	Only One Service	0.829	0.890	−6.9%
Subscription Prices	Netflix	14.678	19.894	−26.2%
(Average $ Per Month)	Amazon Prime	9.546	9.945	−4.0%
	Hulu	8.204		
	Disney Plus	7.829	8.341	−6.1%
Market Shares	Netflix	0.556	0.502	10.6%
(Average Per Month)	Amazon Prime	0.443	0.454	−2.4%
	Hulu	0.289		
	Disney Plus	0.299	0.252	18.4%
	Multi-Homing	0.414	0.279	48.5%
Service Payoffs	Netflix	5.940	6.709	−11.5%
($Bn Per Year)	Amazon Prime	2.416	2.575	−6.2%
	Hulu	0.331		
	Disney Plus	2.056	1.926	6.8%
	Total Service Payoff	10.744	11.210	−4.2%
Studio Payoffs	Big Five	8.470	8.254	2.6%
($Bn Per Year)	Remaining	4.361	4.240	2.9%
	Total Studio Payoff	12.831	12.495	2.7%
Consumer Surplus ($ Per HH-Year)		204.476	183.302	11.6%
Avg. Weekly Streaming Hours		2.537	2.482	2.2%

Notes. Specification (1) simulates the status quo, while (2) removes exclusive contracts and Hulu. The last column reports percentage changes from specification (2) to (1). The share of multi-homing households includes those with at least two subscriptions to Netflix, Amazon Prime, or Hulu (excluding Disney Plus).

Appendix B Details on the Bilateral Contracting Model

B.1 Microfoundation of the Bilateral Contracting Model

Extensive Form Game The extensive form game that yields the bilateral contracting model follows Ho and Lee (2019, page 498). This extensive form game applies to both the simple model in Section 3 and the full empirical model in Section 5. I summarize it and discuss its key intuitions below. Throughout the bargaining process for the licensing right of title j, the participating studio and streaming services assume that all other titles will be distributed as in equilibrium.

In period 0, the studio publicly announces its intended distribution network $\mathcal{K}_j$. For each service $k \in \mathcal{K}_j$, the studio assigns a delegate to negotiate a contract. Each delegate commits to negotiating only with her assigned service k or an excluded service $k' \notin \mathcal{K}_j$. These delegates do not share information about the offers made or received in the subsequent periods.

In periods 1 to T, with T sufficiently large, each studio delegate can choose to engage with its assigned service k or an excluded service k'. This delegate is selected by nature with probability b to make a take-it-or-leave-it (TIOLI) offer to her engaged service, while the engaged service is selected with probability $1 - b$ to propose a TIOLI offer to the delegate. The party that receives the offer can either accept or reject the offer. Once a studio delegate finalizes a contract with a streaming service, she is removed from the game. In period $T + 1$, the payoffs for all parties are realized at the same time.

This framework captures how studios use excluded services as bargaining leverage with two important deviations from the "Nash-in-Nash" solution. First, studios can create licensing right scarcity by credibly excluding some streaming services from distribution. This is achieved by limiting the number of delegates, with each delegate committing to contracting with only one service. In practice, exclusive contracts ensure such commitments. Second, studios can induce competition by allowing delegates to walk away from assigned services and negotiate with excluded ones, forcing streaming platforms to compete for scarce licensing rights.

Equilibrium Outcome In the weak perfect Bayesian equilibrium, the studio selects the most profitable intended network in period 0. If the network is non-exclusive, the outcome corresponds to the standard Nash-in-Nash solution, as microfounded in Collard-Wexler, Gowrisankaran and Lee

(2019). I focus here on deriving the equilibrium license fees when the intended network is exclusive.

I restrict attention to stationary strategies, so that licensing agreements are reached in period 1 in equilibrium. The studio may adopt a mixed strategy to randomize between services k_1 and k_2 with probabilities β_1 and β_2, respectively. All parties share a common discount factor δ. Let Π_1^e and Π_2^e denote joint profits under exclusive contracts with k_1 and k_2, respectively, and assume $\Pi_1^e > \Pi_2^e$ without loss of generality. Let π_j, π_1, and π_2 denote the payoffs to the studio and services k_1 and k_2.

Proposition 1. *When the studio commits to exclusive distribution, its expected payoff converges to* $\max\{b\Pi_1^e, \Pi_2^e\}$ *as* $\delta \to 1$*, and the probability of contracting with* k_1 *converges to 1.*

Proof. This proof builds on and simplifies the proof for Proposition 1 from Manea (2018). For any $k \in \{k_1, k_2\}$ with $\beta_k > 0$, the payoffs under the equilibrium contract are

$$\pi_j = b(\Pi_k^e - \delta\pi_k) + (1-b)\delta\pi_j, \tag{B.1}$$

$$\pi_k = \beta_k \cdot \left[b\delta\pi_k + (1-b)(\Pi_k^e - \delta\pi_j)\right]. \tag{B.2}$$

These payoff equations follow Rubinstein (1982): each player is indifferent between accepting and rejecting, with rejection leading to renegotiation next period. The key difference is that due to the studio's mixed strategy, service k only has probability β_k of being selected in the next round.

Rewrite (π_k, β_k) to be functions of π_j:

$$\pi_k = \frac{\Pi_k^e}{\delta} - \frac{1-\delta+\delta b}{\delta b}\pi_j, \tag{B.3}$$

$$\beta_k = \frac{1-\delta+\delta b}{\delta b} - \frac{(1-\delta)(1-b)\Pi_k^e}{\delta b(\Pi_k^e - \pi_j)}. \tag{B.4}$$

In equilibrium, $\pi_k \geq 0$, otherwise service k is better off by always rejecting any offer from the studio. Therefore, equation (B.3) implies that $\beta_k > 0$ must lead to $\pi_j \leq \frac{b\Pi_k^e}{1-\delta+\delta b}$. In addition, $\beta_k = 0$ must imply $\pi_j > \frac{b\Pi_k^e}{1-\delta+\delta b}$, since with $\beta_k = 0$, there is $u_k = 0$. Therefore, the studio does not negotiate with k at all ($\beta_k = 0$) if it cannot possibly benefit from the negotiations: $\pi_j \geq b\Pi_k^e + (1-b)\delta\pi_j$, or equivalently, $\pi_j > \frac{b\Pi_k^e}{1-\delta+\delta b}$.

Therefore, the studio's equilibrium mixed strategy is:

$$\tilde{\beta}_k(\pi_j) = \begin{cases} \frac{1-\delta+\delta b}{\delta b} - \frac{(1-\delta)(1-b)\Pi_k^e}{\delta b\left(\Pi_k^e - \pi_j\right)} & \text{if } \pi_j < \frac{b\Pi_k^e}{1-\delta+\delta b} \\ 0 & \text{if } \pi_j \geqslant \frac{b\Pi_k^e}{1-\delta+\delta b} \end{cases}. \tag{B.5}$$

I will now discuss the negotiated license fees under the following two scenarios.

Scenario 1: $b\Pi_1^e \geq \Pi_2^e$

Below, I show that $\beta_1 = 1$ for all $\delta \in (0,1)$ if and only if $b\Pi_1^e \geq \Pi_2^e$, with $\pi_j = b\Pi_1^e$. If $\forall\delta \in (0,1)$, $\beta_1 = 1$, equations (B.1) and (B.2) imply that

$$\pi_j = b(\Pi_1^e - \pi_1) + (1-p)\delta\pi_1, \quad \pi_1 = b\delta\pi_1 + (1-b)(\Pi_1^e - \delta\pi_j), \quad \pi_2 = 0. \tag{B.6}$$

Solving this system of equations yields $\pi_j = b\Pi_1^e$ and $\pi_1 = (1-b)\Pi_1^e$ for all $\delta \in (0,1)$. To ensure $\beta_2 = 0$, the mixed strategy function (B.5) implies $\pi_j = b\Pi_1^e \geq \frac{b\Pi_2^e}{1-\delta+\delta b}$ for any $\delta \in (0,1)$. Taking $\delta \to 1$, it implies $b\Pi_1^e \geq \Pi_2^e$.

When $b\Pi_1^e \geq \Pi_2^e$, assuming that $\exists\delta' \in (0,1)$, under which $\beta_1 < 1$ and $\beta_2 > 0$. Equation (B.5) implies that to have $\beta_2 > 0$, the studio's payoff must satisfy $\pi_j \leq \frac{b\Pi_2^e}{1-\delta'+\delta' b} < b\Pi_1^e$, where the last inequality comes from $b\Pi_1^e \geq \Pi_2^e$. However, this strategy is not rational for the studio: it could have achieved $b\Pi_1^e$ by only negotiating with k_1: $\beta_1 = 1$. This contradiction suggests that there must be $\beta_1 = 1$, where the studio earns $b\Pi_1^e$.

Scenario 2: $b\Pi_1^e < \Pi_2^e$

Below, I prove that when $b\Pi_1^e < \Pi_2^e$, there are (a) $\exists\bar{\delta} \in (0,1)$ that under any $\delta > \bar{\delta}$, there is $\beta_2 > 0$; and (b) $\lim_{\delta\to 1}\beta_2 = 0$.

I prove both results by contradiction. (a) Suppose not. Then there exists $\delta' > \delta_0$ such that $\beta_2 = 0$, where δ_0 solves $\frac{b\Pi_2^e}{1-\delta_0+\delta_0 b} = b\Pi_1^e$. With $\beta_2 = 0$, there is $\pi_j = b\Pi_1^e$ under δ'. However, from (B.5), $\pi_j = \frac{b\Pi_2^e}{1-\delta+\delta b} < \frac{b\Pi_2^e}{1-\delta'+\delta' b}$ implies $\beta_2 > 0$, contradicting the assumption. Therefore, (a) holds.

(b) Suppose $\lim_{\delta\to 1}\beta_2 \neq 0$. Since $\beta_1 + \beta_2 = 1$ always holds, $\lim_{\delta\to 1}\beta_2 = \lim_{\delta\to 1}\frac{(1-\delta)(1-b)\Pi_1^e}{\delta b(\Pi_1^e - \pi_j)}$. If $\lim_{\delta\to 1}\beta_2 \neq 0$, there must be $\lim_{\delta\to 1}\pi_j = \Pi_1^e$. However, since $\Pi_1^e > \Pi_2^e$, it means that $\exists\delta_0 \in (0,1)$, such that $\forall\delta \in (\delta_0, 1)$, there is π_j that satisfies $\Pi_1^e \geq \pi_j > \Pi_2^e > \frac{b\Pi_2^e}{1-\delta+\delta b}$, which implies that $\beta_2 = 0$ according to (B.5), a contradiction. Therefore, (b) holds.

Together, (a) and (b) imply that the likelihood for the studio to negotiate with k_2, β_2, is positive but converges to zero with $\delta \to 1$. Combining with (B.5), there is

$$\lim_{\delta \to 1} \pi_j = \lim_{\delta \to 1} \frac{b\Pi_2^e}{1 - \delta + \delta b} = \Pi_2^e. \tag{B.7}$$

Summarizing both scenarios, as $\delta \to 1$, the studio's payoff converges to $\max\{b\Pi_1^e, \Pi_2^e\}$, and the probability of contracting with k_1 approaches 1. □

Key Intuition When $b\Pi_1^e < \Pi_2^e$, the studio benefits from using k_2 as a credible threat to extract better terms from k_1. To make this threat effective, it must assign a non-trivial probability β_2 to bargaining with k_2, especially when δ is low and any bargaining breakdown is costly. Otherwise, if β_2 is too small, k_2 is not an effective threat against k_1, as the studio will have to wait for many periods to select k_2 for bargaining, while this delay is costly for the studio. As δ increases, bargaining breakdowns become less costly, allowing the studio to reduce β_2 while maintaining credibility.

This dynamic also affects k_2's willingness to pay. When δ is small and β_2 remains non-trivial, k_2 knows the studio incurs costs from bargaining breakdowns. As a result, k_2 can retain a small "premium" and pass the rest of Π_2^e to the studio, so that the studio will opt to reach an agreement with k_2 to avoid a breakdown. However, as $\delta \to 1$ and $\beta_2 \to 0$, k_2 realizes it will almost surely not be chosen for bargaining again if its current bargaining with the studio fails. Anticipating zero continuation value, it is willing to surrender nearly all of Π_2^e to secure the deal when selected for bargaining in the current period.

B.2 Simple Model Extension: Heterogeneous Bargaining Power

Below, I extend the simple model from Section 3 to accommodate differential bargaining power among streaming services and show that the core insights from the simple model persist. Moreover, it illustrates how variations in distribution networks can also identify differences in streaming services' bargaining power.

Consider a scenario with one studio, j, and two streaming services, k_1 and k_2, as depicted in Figure 1. Denote the sales profits for k_1 and k_2 under non-exclusive (ne) and exclusive (e) distribution scenarios as Π_1^{ne}, Π_2^{ne} and Π_1^e, Π_2^e, respectively. Without loss of generality, assume

$\Pi_1^e > \Pi_2^e$. The respective bargaining powers of the studio with each streaming service are represented by b_1 and b_2. The analysis follows the game timeline and equilibrium concept outlined in Section 3.

The NNTR bargaining solution implies the payoffs for the studio under exclusive ($\mathcal{K}_j = \{k_1\}$) and non-exclusive ($\mathcal{K}_j = \{k_1, k_2\}$) networks as follows:

$$\Pi_j(\mathcal{K}_j) = \sum_{k \in \mathcal{K}_j} \tau_{jk}(\mathcal{K}_j) = \begin{cases} \max\{b_1\Pi_1^e, \Pi_2^e\}, & \mathcal{K}_j = \{k_1\} \\ b_1\Pi_1^{ne} + b_2\Pi_2^{ne}, & \mathcal{K}_j = \{k_1, k_2\} \end{cases}. \tag{B.8}$$

The studio will opt for a non-exclusive distribution strategy if and only if it yields a higher payoff, which requires the satisfaction of the following conditions:

$$b_2\Pi_2^{ne} \geq \Pi_2^e - b_1\Pi_1^{ne}, \tag{B.9}$$

$$(\Pi_1^{ne} + \Pi_2^{ne}) + \left(\frac{b_2}{b_1} - 1\right)\Pi_2^{ne} \geq \Pi_1^e. \tag{B.10}$$

Condition (B.9) mirrors (5), reflecting the studio's preference for using exclusive contracts as leverage. If the extra surplus from k_2 when including it in a non-exclusive network (left-hand side of the inequality), exceeds the reduction in surplus from k_1 (right-hand side), the studio will choose a non-exclusive distribution network.

Condition (B.10) is different from (4) due to varying bargaining powers of the streaming services, encapsulated in the term $\left(\frac{b_2}{b_1} - 1\right)\Pi_2^{ne}$. This term reflects how differential bargaining powers influence the studio's preferences: a higher b_2 compared to b_1 suggests that the studio is able to extract more surplus from k_2, thereby valuing k_2's sales profit more. Conversely, when b_1 surpasses b_2, the studio prioritizes k_1's sales profit. Thus, this term reflects the studio's steering incentive based on the relative bargaining power of the streaming services.

In addition, condition (B.10) affirms the significance of network efficiency, which is also suggested by (4). The efficiency of a non-exclusive network (measured by $\Pi_1^{ne} + \Pi_2^{ne}$) over an exclusive one (Π_1^e) increases the likelihood of the studio choosing the former. In sum, (B.9) and (B.10) highlight the balance between leveraging bargaining power and optimizing network efficiency in the studio's distribution strategy.

Identification of Differential Bargaining Powers of Streaming Services The necessary and sufficient condition for the studio to opt for a non-exclusive network, $\Pi_j(\{k_1\}) \leq \Pi_j(\{k_1, k_2\})$, can be rewritten as

$$b_2 \leq \frac{\max\{b_1\Pi_1^e, \Pi_2^e\} - b_1\Pi_1^{ne}}{\Pi_2^{ne}}. \tag{B.11}$$

This condition illustrates that, all else being equal, a studio is more inclined to exclude service k_2 from the network when k_2 has stronger bargaining power (indicated by a smaller b_2). Intuitively, the studio prefers to add k_2 into the network when it can extract substantial surplus from it.

The interaction between distribution and k_1's bargaining power, b_1, involves two countervailing forces. A larger b_1 implies the studio has strong bargaining power against k_1, reducing the need to exclude k_2 as leverage when bargaining with k_1. However, a larger b_1 also means the studio can extract more surplus from k_1, motivating the studio to enhance k_1's profitability by providing exclusive rights to k_1. Therefore, the effect of b_1 on exclusive distribution depends on the difference between k_1's subscription profits under exclusive and non-exclusive distributions: $\Pi_1^e - \Pi_1^{ne}$. When this difference is small, a large b_1 implies a reduced likelihood of exclusive distribution and using k_2 as a threat against k_1. In contrast, when such difference is large, a large b_1 implies a strong likelihood of excluding k_2 to guard the profit of k_1 that the studio can attract from licensing contracts. In short, the covariance between $\Pi_1^e - \Pi_1^{ne}$ and exclusive distribution can identify b_1.

This analysis shows that differences in bargaining powers among streaming services can be identified through variation in title distribution. While the main specification assumes uniform bargaining power across streaming services for simplicity, I explore additional specifications in Appendix E.2, where bargaining power varies across streaming services.

B.3 NNTR Solution with Lump-sum Licensing Fees

Non-Integrated Bargaining I start by discussing the scenario where the studio and streaming service are not vertically integrated with each other. When the "threat of replacement" constraint (18) is not binding, the first-order condition of Nash product maximization (17) implies that

$$(1 - b_{jk}) \cdot \Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}^{NN}, \boldsymbol{\tau}_{-jk}\}) = b_{jk} \cdot \Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}^{NN}, \boldsymbol{\tau}_{-jk}\}). \tag{B.12}$$

Given that licensing fees are lump-sum payments and do not impact consumer demand or streaming service pricing conditional on distribution networks, the gains-from-trade can be expressed as

$$
\begin{aligned}
\Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}^{NN}, \boldsymbol{\tau}_{-jk}\}) &= \Delta_{jk}\Pi_j(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) + \tau_{jk}^{NN}, \\
\Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}^{NN}, \boldsymbol{\tau}_{-jk}\}) &= \Delta_{jk}\Pi_k(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) - \tau_{jk}^{NN}.
\end{aligned}
\tag{B.13}
$$

Consequently, the first-order condition above can be reconstructed as

$$
\tau_{jk}^{NN} = b_{jk} \cdot \Delta_{jk}\Pi_k(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) - (1 - b_{jk}) \cdot \Delta_{jk}\Pi_j(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}). \tag{B.14}
$$

Under this solution, the studio's gain-from-trade with service k is:

$$
\Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}^{NN}, \boldsymbol{\tau}_{-jk}\}) = b_{jk} \cdot \Delta_{jk}\Pi_{jk}(\mathcal{K}_j), \tag{B.15}
$$

where $\Delta_{jk}\Pi_{jk}(\mathcal{K}_j) = \Delta_{jk}\Pi_j(\mathcal{K}_j, \cdot) + \Delta_{jk}\Pi_k(\mathcal{K}_j, \cdot)$ represents the bilateral surplus from the licensing agreement between the two firms. Notably, this bilateral surplus is not a function of the negotiated licensing fees because they are lump sums.

When constraint (18) is binding, the reservation licensing fee of excluded service k' is defined to satisfy

$$
\tau_{jk'}^{res} = \Delta_{jk'}\Pi_{k'}((\mathcal{K}_j \setminus k) \cup k', \{0, \boldsymbol{\tau}_{-jk}\}), \tag{B.16}
$$

which implies that k' is willing to surrender all bilateral surplus to the studio if the latter chooses to switch from k to k'. Therefore, the "threat of replacement" constraint implies that the negotiated licensing fee must satisfy the following condition:

$$
\Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}^{TR}, \boldsymbol{\tau}_{-jk}\}) = \max_{k' \notin \mathcal{K}_j} \left\{ \Delta_{jk'}\Pi_{jk'}((\mathcal{K}_j \setminus k) \cup k', \{0, \boldsymbol{\tau}_{-jk}\}) \right\}. \tag{B.17}
$$

Combining (B.15) and (B.17), under the NNTR solution $\tau_{jk}^* = \max\{\tau_{jk}^{NN}, \tau_{jk}^{TR}\}$, the studio's gain-from-trade with a contracting partner $k \in \mathcal{K}_j$ corresponds to equation (21):

$$
\Delta_{jk}\Pi_j(\mathcal{K}_j, \boldsymbol{\tau}^*) = \max \left\{ b_{jk} \cdot \Delta_{jk}\Pi_{jk}(\mathcal{K}_j), \max_{k' \notin \mathcal{K}_j} \left[\Delta_{jk'}\Pi_{jk'} \left((\mathcal{K}_j \setminus k) \cup k' \right) \right] \right\}. \tag{21}
$$

Licensing Fees Between Vertically Integrated Firms I then discuss the scenario where a studio bargains with its vertically integrated streaming service. For simplicity, this analysis assumes $\mu < 1$, indicating incomplete vertical integration. The validity of the assumption is further explored at the end of this section. When the "threat of replacement" constraint is not binding, the first-order condition of Nash-in-Nash bargaining leads to

$$\begin{aligned} &(1 - b_{jk}) \cdot \left[\Delta_{jk}\pi_j(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) + \tau_{jk}^{NN} + \mu \cdot \left(\Delta_{jk}\Pi_k(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) - \tau_{jk}^{NN}\right)\right] \\ =& b_{jk} \cdot \left[\Delta_{jk}\Pi_k(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) - \tau_{jk}^{NN}\right], \end{aligned} \tag{B.18}$$

where π_j represents the studio's payoff, excluding effects from its vertically integrated counterparts, and is defined as $\pi_j = \sum_{k \in \mathcal{K}_j}[\tau_{jk} + \nu_{jk}(\mathcal{K}_j)] + \gamma \cdot V_j(\mathcal{K}_j)$. This equation determines the licensing fee under non-binding constraints as

$$\tau_{jk}^{NN} = -\frac{(1-b) \cdot \Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \cdot)}{1 - \mu(1-b)} + \Delta_{jk}\Pi_k(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}), \tag{B.19}$$

where $\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j) = \Delta_{jk}\pi_j(\mathcal{K}_j, \cdot) + \Delta_{jk}\Pi_k(\mathcal{K}_j, \cdot)$ is the maximum gain-from-trade the studio could secure when licensing to its vertically integrated streaming service, given that the service surrenders all its surplus to the studio. It is also not a function of τ.

When the "threat of replacement" constraint is binding, the studio's gain-from-trade with service $k \in \mathcal{K}_j$ must equal or surpass the bilateral surplus from licensing to any excluded service $k' \notin \mathcal{K}_j$:

$$\begin{aligned} &\Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}^{TR}, \boldsymbol{\tau}_{-jk}\}) = \Delta_{jk}\pi_j(\mathcal{K}_j, \{\tau_{jk}^{TR}, \boldsymbol{\tau}_{-jk}\}) + \mu \cdot \Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}^{TR}, \boldsymbol{\tau}_{-jk}\}) \\ =& \max_{k' \notin \mathcal{K}_j} \left[\Delta_{jk'}\tilde{\Pi}_{jk'}\left((\mathcal{K}_j \setminus k) \cup k'\right) + \mu \cdot \Delta_{jk'}\Pi_k\left((\mathcal{K}_j \setminus k) \cup k', \cdot\}\right)\right]. \end{aligned} \tag{B.20}$$

Notably, $\Pi_k(\mathcal{K}_j \setminus k \cup k', \cdot)$ remains constant irrespective of lump-sum transfers between service k' and the studio because any lump-sum transfer between competing service k' and the studio does not affect the pricing nor consumer demand of k, conditional on the distribution network. The negotiated licensing fee under a binding constraint becomes:

$$\begin{aligned}\tau_{jk}^{TR} =& \frac{1}{1-\mu}\left(\max_{k'\notin\mathcal{K}_j}\left[\Delta_{jk'}\tilde{\Pi}_{jk'}\left((\mathcal{K}_j\setminus k)\cup k'\right)+\mu\cdot\Delta_{jk'}\Pi_k\left((\mathcal{K}_j\setminus k)\cup k',\cdot\right)\right]-\Delta_{jk}\tilde{\Pi}_{jk}\left(\mathcal{K}_j\right)\right)\\ &+\Delta_{jk}\Pi_k(\mathcal{K}_j,\{0,\boldsymbol{\tau}_{-jk}\}).\end{aligned} \tag{B.21}$$

The term in the first line must be non-positive following the stability condition 22, which mandates that the bilateral surplus between the studio and any excluded service k' (denoted within the brackets) must not exceed the maximum possible gain-from-trade that the studio can obtain from service k, $\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j,\cdot)$.

Under the NNTR solution, the negotiated licensing fee between two vertically integrated firms can be represented as $\tau_{jk}=\max\{\tau_{jk}^{NN},\tau_{jk}^{TR}\}$. Notably, τ_{jk} decreases with μ when $\mu<1$, as both τ_{jk}^{NN} and τ_{jk}^{TR} do. Subsequently, the gain-from-trade perceived by the studio from contracting with its vertically-integrated service k is:

$$\begin{aligned}\Delta_{jk}\Pi_j(\mathcal{K}_j,\boldsymbol{\tau}^*)=\max&\left\{\frac{b}{1-(1-b)\mu}\cdot\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j,\{0,\boldsymbol{\tau}_{-jk}\}),\right.\\ &\left.\max_{k'\notin\mathcal{K}_j}\left[\Delta_{jk'}\tilde{\Pi}_{jk'}((\mathcal{K}_j\setminus k)\cup k',\cdot)+\mu\cdot\Delta_{jk'}\Pi_k((\mathcal{K}_j\setminus k)\cup k',\cdot)\right]\right\}\end{aligned} \tag{B.22}$$

Discussion: Assumption of $\mu<1$ The above analysis adopts the assumption that $\mu<1$. By contrast, $\mu\geq 1$ implies that a studio prioritizes the payoff to its vertically integrated service over its own, so it would accept a licensing fee approaching negative infinity. This scenario is not consistent with actual market behaviors, where Hulu pays licensing fees to studios affiliated with Walt Disney. Moreover, the assumed $\mu<1$ aligns with documented internal frictions within conglomerates (e.g., Crawford et al. 2018, Hortaçsu et al. 2024).

B.4 Distribution Network Formation: The Stability Condition

In this section, I derive the sufficient and necessary condition of the stability condition, denoted as $\Delta_{jk}\Pi_k(\mathcal{K}_j,\{\tau_{jk},\boldsymbol{\tau}_{-jk}\})\geq 0$, for non-integrated and vertically integrated firms in Proposition 2 and 3, respectively.

Proposition 2. *The stability condition for two non-integrated firms, j and k, is equivalent to*

$$\Delta_{jk}\Pi_{jk}(\mathcal{K}_j) \geq \Delta_{jk'}\Pi_{jk'}\left((\mathcal{K}_j \setminus k) \cup k'\right), \forall k' \notin \mathcal{K}_j, \tag{22}$$

where $\Delta_{jk}\Pi_{jk}(\mathcal{K}_j)$ represents the bilateral surplus generated by j and k reaching an agreement with each other.

Proof. This proposition resembles Proposition 2 from Ho and Lee (2019). Under distribution network $\mathcal{K}_j$ and NNTR solution $\boldsymbol{\tau}^*$, the gain-from-trade for non-integrated service k from licensing title j is

$$\begin{aligned}&\Delta_{jk}\Pi_k(\mathcal{K}_j, \boldsymbol{\tau}^*) = \Delta_{jk}\Pi_{jk}(\mathcal{K}_j) - \Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}, \boldsymbol{\tau}_{-jk}\})\\ &= \min\left\{(1-b_{jk}) \cdot \Delta_{jk}\Pi_{jk}(\mathcal{K}_j), \Delta_{jk}\Pi_{jk}(\mathcal{K}_j) - \max_{k' \notin \mathcal{K}_j}\left[\Delta_{jk'}\Pi_{jk'}\left((\mathcal{K}_j \setminus k) \cup k'\right)\right]\right\}\end{aligned} \tag{B.23}$$

Here, the first equality is derived from the definition of bilateral surplus and its independence from negotiated lump-sum licensing fees, while the second equality is obtained by substituting the studio's gain-from-trade $\Delta_{jk}\Pi_j(\mathcal{K}_j, \tau_{jk}, \boldsymbol{\tau}_{-jk})$ with equation (21). Considering that the bilateral surplus is always positive in this study, the stability condition can be reformulated as (22). □

Proposition 3. *Assuming the internalization parameter $\mu < 1$, the stability condition for two vertically integrated firms, j and k, is equivalent to*

$$\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j) \geq \Delta_{jk'}\tilde{\Pi}_{jk'}\left((\mathcal{K}_j \setminus k) \cup k'\right) + \mu \cdot \Delta_{jk'}\Pi_k\left((\mathcal{K}_j \setminus k) \cup k'\right), \forall k' \notin \mathcal{K}_j, \tag{B.24}$$

where $\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \cdot) = \Delta_{jk}\pi_j(\mathcal{K}_j, \cdot) + \Delta_{jk}\Pi_k(\mathcal{K}_j, \cdot)$.

Proof. For service k licensing title j from its vertically integrated studio, its gain-from-trade, denoted as $\Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}, \boldsymbol{\tau}_{-jk}\})$, has the following property:

$$\begin{aligned}&(1-\mu) \cdot \Delta_{jk}\Pi_k(\mathcal{K}_j, \{\tau_{jk}, \boldsymbol{\tau}_{-jk}\}) = \Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \cdot) - \Delta_{jk}\Pi_j(\mathcal{K}_j, \{\tau_{jk}, \boldsymbol{\tau}_{-jk}\})\\ &= \min\left\{\frac{(1-b)(1-\mu)}{1-(1-b)\mu} \cdot \Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}),\right.\\ &\qquad \left.\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \cdot) - \max_{k' \notin \mathcal{K}_j}\left[\Delta_{jk'}\tilde{\Pi}_{jk'}((\mathcal{K}_j \setminus k) \cup k', \cdot) + \mu \cdot \Delta_{jk'}\Pi_k((\mathcal{K}_j \setminus k) \cup k', \cdot)\right]\right\},\end{aligned} \tag{B.25}$$

where the first equality is derived from the definition of $\tilde{\Pi}_{jk}(\mathcal{K}_j, \cdot)$, while the second is derived from substituting the term in the bracket with its equivalence from (B.22).

Because $\mu < 1$ and $\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) > 0$, $\frac{(1-b)(1-\mu)}{1-(1-b)\mu} \cdot \Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j, \{0, \boldsymbol{\tau}_{-jk}\}) > 0$ always holds. Consequently, the stability condition can be translated to

$$\Delta_{jk}\tilde{\Pi}_{jk}(\mathcal{K}_j) - \max_{k' \notin \mathcal{K}_j} \left[\Delta_{jk'}\tilde{\Pi}_{jk'}((\mathcal{K}_j \setminus k) \cup k') + \mu \cdot \Delta_{jk'}\Pi_k((\mathcal{K}_j \setminus k) \cup k')\right] \geq 0. \tag{B.26}$$

which is equivalent to condition (B.24). $\square$

Proposition 3 shows that vertical integration increases the likelihood of a studio finding distribution to its vertically integrated streaming service stable. While vertical integration does not increase the maximum gain-from-trade the studio can extract from a streaming service, $\Delta\tilde{\Pi}_{jk}(\mathcal{K}_j)$, as this term is independent of μ, it reduces the perceived benefit of licensing to competitors. This is because such contracts harm the vertically integrated service's payoff, which the studio considers when distributing titles.

Appendix C Details on Data

C.1 Details on Nielsen Ratings Data

I use Nielsen ratings data from March 2021 to February 2022 in this paper. The data are at the weekly level and also offer ratings by demographic breakdowns, including age groups (2–17, 18–44, and 45+ years old), gender (male and female), and race (white, African American, and others).

Nielsen measures title viewership by monitoring and surveying households within their panel. Two types of information are collected. The first type concerns which titles are watched by each household on any of their screens. This information is collected using Automatic Content Recognition (ACR), a pattern-matching technology. ACR collects short audio and visual clips of media played on screens, which are then cross-referenced with a library of signals from shows and movies to identify the watched content. Viewership of a title by an individual is accounted only if they meet specific criteria, such as a minimum continuous viewing of fifteen minutes.

The second type of data is the demographics of the viewers. Nielsen conducts in-house surveys to collect demographic information and validate the subscription choices of surveyed households simultaneously. These surveys are conducted regularly, typically every six months. In addition, Nielsen installs meters in the households within their survey panel to identify the viewer every time a screen is on. Household members are required to report their unique survey ID upon turning on a screen to confirm their identity.

C.2 Sample and Variable Construction

In this section, I outline the methodology for constructing variables on market shares, subscription prices, and title characteristics. I also detail the criteria for selecting titles in the final sample and discuss their implications.

Prices Netflix and Hulu provide multiple subscription tiers. To calculate their average consumer prices, I interviewed experts from these companies and data analysts from YipitData, a data vendor that collects SVOD subscription receipts from millions of U.S. users' mailboxes. Both sources provided tier-wise subscriber distributions, and their data closely matched. The average prices for Netflix and Hulu were then computed by averaging the prices across different tiers, weighted by the proportion of subscribers in each tier. To validate these calculations, I compared the derived average prices for Netflix with those reported in Netflix's financial statements. The difference between these two sources was minimal, with discrepancies consistently under \$0.5 for all quarters.

Market Shares Market shares for each bundle of streaming services, defined as a unique combination of the top four services, are sourced from Nielsen Household Universe Estimates. However, there is a potential for overestimation in the Nielsen data. For example, Nielsen reports 77 million subscribers in the U.S. for Netflix in March 2021, while Netflix's fiscal report has only 74 million subscribers across both the U.S. and Canada. This discrepancy can result from two factors. The first is shared subscription accounts across households. Nielsen usually identifies a household's subscription by checking if any member can browse titles on each service through installed meters. In addition, Nielsen conducts in-house surveys to detect account sharing and to ensure the precision of their subscriber numbers. However, undetected password sharing could still occur, potentially leading to an overcount of subscribers. The second reason is differing definitions of subscriptions. Fiscal reports from streaming services typically count subscribers as of a specific date, whereas

Nielsen considers a household a subscriber if they have had access to a service at any point within a month.

To rectify this, I first calculate the ratio of national subscribers according to Nielsen versus those reported in fiscal documents for each of the top four streaming services over 12 months. Since Netflix combines its U.S. and Canadian subscriber numbers, I assume that both countries have an equal market share when calculating this ratio. I then apply this ratio to adjust the number of subscribers for all streaming service *bundles* that include these services across all DMAs and months. As a result, the final sample's total household subscription numbers for each of the top four services should more accurately reflect the figures reported in fiscal documents.

Title Sample Selection. The original dataset from Nielsen ratings consists of 8,835 titles, which Nielsen can detect viewership from their sampled households. In the paper, the final sample is narrowed down to the top 2,028 titles. They are selected based on their maximum or average weekly rating, which must fall within the top 20% for at least one of the streaming services to which they are distributed.

A limitation of this selection criterion is its potential bias towards including non-exclusive titles in the final sample. This problem arises because Nielsen ratings do not provide separate viewership data for non-exclusive titles by the services that viewers are from. Consequently, non-exclusive titles are more likely to have higher ratings than exclusive ones. This discrepancy leads to a challenge to the identification of bargaining parameters, which relies on the likelihood of studios opting for exclusive distribution.

While I cannot entirely eliminate this concern, the composition of the final sample offers reassurance. 86.7% of the sampled third-party titles opt for exclusive distribution, closely aligning with 86.1% observed among all titles available on the top four services, according to Reelgood data. This similarity supports the representativeness of the selected sample.

Title Characteristics Variables. The age of a title is constructed based on the number of weeks elapsed since its most recent release or update. This means that the age of a title resets to one week whenever a new episode is released. Titles that have not been released or updated in the last 50 weeks are categorized as "old," and their age is marked as zero in the dataset. In addition, a show is classified as binge-released if, during its latest release or update, it debuts at least four episodes.

The production of most titles involves multiple studios. For these titles, the studio holding the most market power is often in charge of their distributions. Therefore, if a title's production includes any of the "Big Five" studios, I assume that studio leads the contract negotiations with streaming services, and the title is assigned as a "Big Five" title.

Appendix D Details on Demand Estimation

D.1 Computational Algorithm

The parameters to estimate are $\theta_1 = \{\boldsymbol{\alpha}^{\mathrm{x}}, \bar{\beta}^0, \bar{\boldsymbol{\beta}}^{\mathrm{w}}\}$ and $\theta_2 = \{\alpha^V, \bar{\alpha}^p, \alpha^p_{inc}, \beta^0_d, \boldsymbol{\beta}^{\mathrm{w}}_d, \beta^0_v, \kappa\}$. θ_1 includes coefficients that are homogeneous for all households and individuals. θ_2 includes all nonlinear parameters in the utility functions, as well as α^V and $\bar{\alpha}^p$ since content utilities and prices vary across households within the same market. The estimation follows the nested fixed point approach from Berry, Levinsohn and Pakes (1995). The outer loop searches for θ_2 that minimizes the GMM objective function, while the inner loop equates simulated and observed demand for both service subscriptions and titles, solves for θ_1, and evaluates the objective function at a given θ_2.

Here are the details. I first recover ζ_{jt} and ξ_{cm} following the iterative contraction mapping method from Lee (2013). I illustrate the approach in Figure D.1. For each guess of θ_2, I obtain an initial estimate of the content utility V_{hcm} for each combination of household, service bundle, and market, by assuming homogeneous preferences for all households and individuals. Next, using the contraction mapping introduced by Berry, Levinsohn and Pakes (1995), I recover the mean service bundle utility in each market ($\delta^S_{cm} = \mathbf{x}_{cm}\boldsymbol{\alpha}^{\mathrm{x}} + \xi_{cm}$). The recovered δ^S_{cm} must correspond to the predicted service bundle shares under the given guess of θ_2 and V_{hcm}. I then calculate the probabilities for each household to select a bundle using the recovered δ^S_{cm}, which are used to estimate the title viewing decisions for each individual. The contraction mapping process for mean title utilities ($\delta^T_{jt} = \mathbf{w}_{jt}\bar{\boldsymbol{\beta}}^w + \zeta_{jt}$) is similar to that for service bundle subscription choices. After δ^T_{jt} converges, the content value variables (denoted as V^+_{hcm}) are updated using definition (12).

The procedure iterates between the contraction mappings of δ^S_{cm} and δ^T_{jt} until V_{hcm} converges. Once convergence is achieved, I project δ^S_{cm} and δ^T_{jt} on $\mathbf{x}_{cm}$ and $\mathbf{w}_{jt}$, respectively, to compute the unobserved service bundle and title demand shocks, ξ_{km} and ζ_{jt} and evaluate the GMM objective.

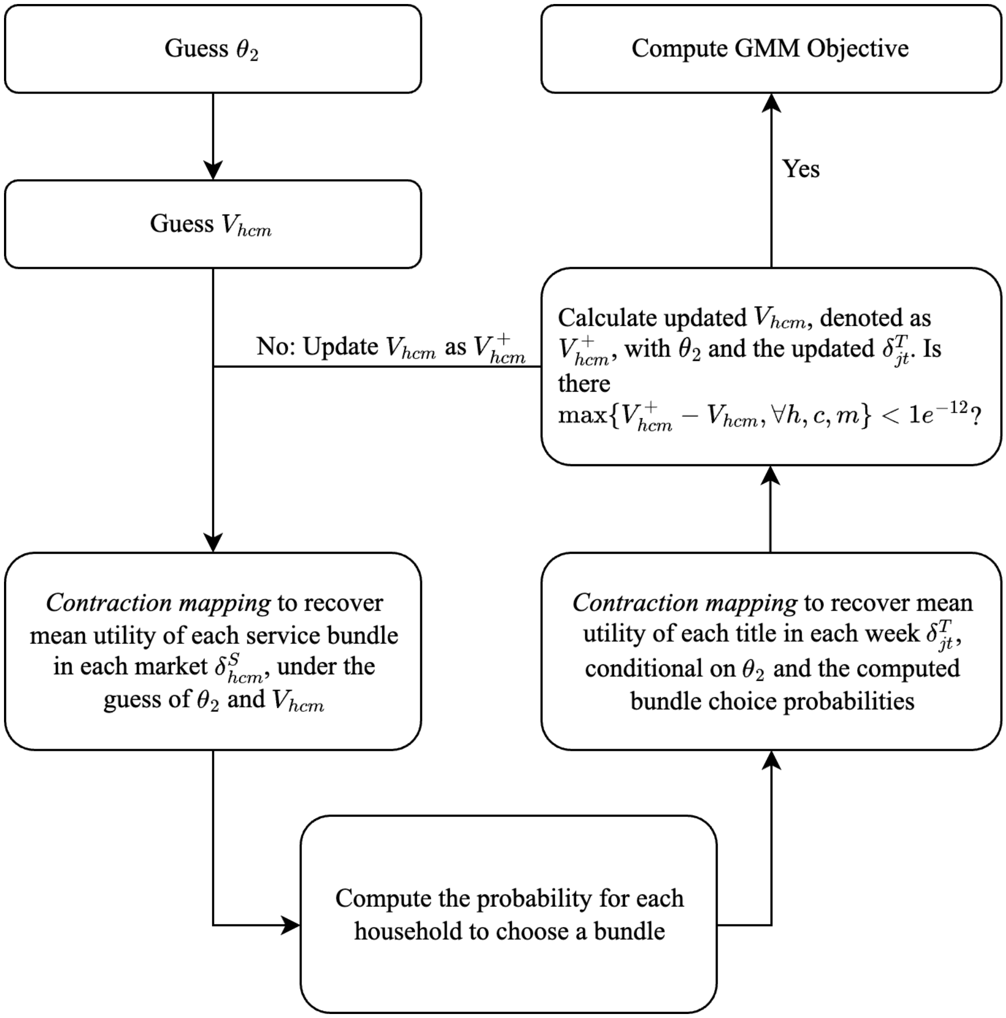


Figure D.1: Nested Fixed Point Algorithm

Using multiple starting values, I find the vector that minimizes the objective function, which serves as the estimate of θ_2.

D.2 Sampling Errors in Integration

The contraction mapping procedure described in D.1 involves simulating market shares of bundle services and titles. Therefore, it requires integration of choice probabilities over different household and individual types, including random coefficients $\{\alpha^p_h, \beta^0_i, \boldsymbol{\beta}^g_i\}$, and household compositions.

Typically, Monte Carlo and Gaussian quadrature are used for such integrations. However, both present challenges for this project. First, the Monte Carlo method is prone to significant sampling errors with "heavy tails," which is critical here as over 8% of households in the top 30 DMAs consist of five or more members, who are expected to show stronger subscription demand. Furthermore, both methods are computationally inefficient: Monte Carlo requires calculating the probability of each title choice for every sampled individual in simulating title market shares (as shown in Figure D.1), a process hindered by the variation in unobserved demographics v_i from equation (12). Though Gaussian quadrature is somewhat more efficient for simulating title market shares, its use in simulating bundle service market shares becomes prohibitively time-consuming: a household of seven members needs $5^7 > 70,000$ nodes for an accuracy level of four.

To address these challenges, I employ the sparse grid method from **?** for integrating over unobserved demographics $v_i^0, \forall i \in h$. This method extends the Gaussian quadrature approach but avoids the curse of dimensionality. The key idea is that, to achieve a certain level of accuracy k, researchers only need quadrature rules that are exact for polynomials of order $2k - 1$ or lower. It outperforms the Monte Carlo method in accuracy and is significantly more computationally efficient than Gaussian quadrature for the purposes of this project.

Simulation-Based Numerical Integration with Sparse Grid Method. I start with randomly selecting households from each of the 30 DMAs, based on 2020 census data, with an upper limit of seven members per household. Any household larger than seven is replaced with a randomly selected seven-member household. For these 3000 sampled households, I apply a sparse grid with an accuracy level of four. This results in 116,514 households, which consist of 470,489 individuals. The individuals can be categorized into 126 demographic groups defined by both observed and unobserved demographics.

Appendix E Details on Supply Estimation

E.1 Computational Details: Supply

Step 1: Recovering r_k The process of recovering the per-subscriber revenue beyond subscriptions for streaming services, r_k, relies on the first-order condition of optimal pricing (15). This step involves calculating the expected sales profits of streaming services and their derivatives with respect to subscription prices. To assess the expected profits and their derivatives, I employ 25 random draws of (ζ, ξ) from the empirical distribution obtained from demand estimation. Specifically, the demand shocks for titles, ζ, are drawn based on the age and type (movie or TV show) of the titles. This is because the variation in ζ tends to decrease as a title ages and that the variation in ζ for

TV shows is generally larger, due to the increasing knowledge of title quality over time and a better understanding of movie quality compared to show quality. The mean values of these simulated sales profits and the derivatives of subscription prices across all 25 draws are then employed to recover the "benefit" per consumer.

Step 2: Simulated Method of Moments Under each guess of bargaining parameters and studio payoff parameters, I simulate the probability of each distribution network to be the equilibrium outcome for every title, $\hat{P}$. This requires evaluating the expected market shares of streaming services and titles under each network. Using 25 random draws of demand shocks (ζ, ξ), I calculate expected market shares and title viewership while keeping other titles' distributions and subscription prices constant, consistent with the assumption that all titles' bilateral contracting and subscription price setting occur simultaneously.

I also use 1000 random draws of ν. For each draw of ν, I determine the equilibrium distribution network for each title that satisfies both the stability condition (22) and the optimality condition (23), holding subscription prices and other titles' distribution networks fixed. The network yielding the highest studio payoff is selected as the equilibrium. The simulated probability of each network being an equilibrium across all draws is used as $\hat{P}$.

To ensure robust optimization and avoid local minima, I use the Matlab command fminsearchOS, an improved version of the Nelde-Meade search algorithm. It is more effective in getting over the kinks and identifying the global minimum. I also use 10 different starting points to ensure the convergence to the global minimum.

E.2 Robustness Checks

Identifying Profit Margins of Prime Video Using Title Distribution Variation In the main model, I estimate the profit margin of Amazon Prime Video using first-order conditions (15). However, this margin can be overestimated if many Prime Video users are existing Prime members who would retain their full Prime subscription regardless of Prime Video, or underestimated if many Prime Video users subscribe to full Prime membership mainly for Prime Video and would cancel their subscription if it lacked sufficient content.

To test the accuracy of this estimation, I re-specify Prime Video's profit margin as $\phi \cdot \hat{\eta}$, where

$\hat{\eta}$ is the margin, or $p_k + r_k$, recovered from first-order conditions (15). I then estimate ϕ using variation in title distribution. A $\phi > 1$ would indicate an underestimation of Prime Video's margins in the main model, while $\phi < 1$ would suggest an overestimation.

The identification of ϕ relies on variation in title distributions. If $\phi < 1$, some titles predicted to be exclusively distributed to Amazon based on the recovered margins $\hat{\eta}$ may fail to satisfy the stability condition (22), meaning they would not be observed to be exclusively available on Amazon. In contrast, if $\phi > 1$, the opposite would hold. In short, the difference between the predicted and observed likelihood of exclusive distribution helps identify ϕ.

I estimate this Amazon payoff parameter ϕ together with other supply-side parameters $\boldsymbol{b}, \gamma, \sigma_\nu, \mu$ using the same moment conditions as in the main model. I report the results in Table E.1. The estimated value of ϕ is 0.98 with a small standard error (0.04), showing strong support for the profit margins recovered from first-order conditions.

Table E.1: Supply Estimation: With Amazon Payoff Parameter

	Estimates	SE
Bargaining Parameters $\boldsymbol{b}$		
"Big Five"	0.790	0.102
Smaller Studios	0.463	0.244
Studio Payoff Parameters		
Logged Viewership γ	0.827	0.222
STD of Unobserved Preferences σ_ν	0.190	0.099
Internalization μ	0.635	0.130
Amazon Payoff Parameter ϕ	0.980	0.036

Notes. Studios' payoffs are measured in millions of dollars. Standard errors are computed using 100 bootstrap samples.

Alternative Studio Categorization The production of most titles involves collaboration between multiple studios. In the main specification, I categorize titles into two groups based on whether any of their production studios belong to the "Big Five." This categorization reflects the industry norm that if at least one studio is among the "Big Five," it typically handles contract negotiations due to its strong bargaining power.

To ensure the robustness of the estimation results to different studio categorizations, I re-estimate the model using an alternative categorization approach. Specifically, I divide titles involving at least one "Big Five" studio into two categories: those where at least 50% of the production companies are "Big Five," and those where less than 50% are "Big Five." This results in 216 titles in the first category, named "Big Five: Major," and 327 titles in the second category, named "Big Five: Minor."

I re-estimate the supply model, allowing for different bargaining powers for these two categories. The regression results, reported in Table E.2, show that the parameters change only marginally from the main specification in Table 5. Specifically, the bargaining powers of studios for these two categories are indifferent, reflecting the industry norm of larger studios handling contract negotiations and supporting the validity of the main specification.

Table E.2: Supply Estimation: Alternative Studio Categorization

	Estimates	SE
Bargaining Parameters $\boldsymbol{b}$		
Big Five: Major	0.817	0.155
Big Five: Minor	0.820	0.117
Smaller Studios	0.576	0.152
Studio Payoff Parameters		
Logged Viewership γ	0.869	0.207
STD of Unobserved Preferences σ_ν	0.164	0.030
Internalization μ	0.657	0.134

Notes. Studios' payoffs are measured in millions of dollars. Standard errors are computed using 100 bootstrap samples.

Differential Bargaining Powers of Streaming Services In the main specification, I assume that all streaming services have the same bargaining powers against studios. As a robustness check, I relax this assumption by allowing Netflix, the largest streaming service worldwide, to have a different bargaining power from its competitors, Amazon Prime and Hulu. I parameterize the bargaining power as follows:

$$b_{jk} = \mathbf{1}(j \in \{\text{Smaller Studios}\})b_{small} + \mathbf{1}(j \in \{\text{Big Five}\})b_{big} - \mathbf{1}(k = \text{Netflix})b_{Netflix}. \tag{E.1}$$

The parameter $b_{Netflix}$ captures the difference in bargaining power between Netflix and its competitors. A larger $b_{Netflix}$ implies stronger bargaining power for Netflix.

The identification of $b_{Netflix}$ relies on the variation in distribution networks. As shown in Section B.2, all else equal, a streaming service with stronger bargaining power is more likely to be excluded, as the studio suffers smaller losses from excluding it.

I present the estimation results in able E.3. I find that the difference in bargaining powers between Netflix and its competitors is statistically insignificant and small in magnitude. This robustness check suggests that the main specification has captured most of the variation in bargaining powers across pairs of firms.

Table E.3: Supply Estimation: Differential Bargaining Power for Services

	Estimates	SE
Bargaining Parameters *b*		
"Big Five"	0.866	0.158
Smaller Studios	0.559	0.154
Against Netflix	−0.063	0.169
Studio Payoff Parameters		
Logged Viewership γ	0.780	0.187
STD of Unobserved Preferences σ_ν	0.146	0.026
Internalization μ	0.626	0.124

Notes. Studios' payoffs are measured in millions of dollars. Standard errors are computed using 100 bootstrap samples.

Limiting the Sample to Newly Released Titles I conduct this robustness check to test if the estimation results are robust to the static bargaining assumption. In addition to the defense of this assumption in Section 5.3, I limit the sample to include only titles released during the observation period. Unlike older titles, studios that produce these new titles should not face any switching costs when choosing the contractual streaming services, even if such costs exist. If switching costs were of first-order importance, the estimates based on this more restricted sample would be different from those based on the full sample.

In Table E.4, I present the estimation results using this restricted sample. I find that all parameters are not statistically different from those using the full sample, including the bargaining parameters, though the standard errors become larger due to the smaller sample size ($N = 244$) compared to the full sample ($N = 1145$).

E.3 In-Sample Model Fit

To investigate in-sample model fit, I simulate the equilibrium outcomes using the model and estimates, following the algorithm described in Section 8.1. In the first two columns of Table E.5, I present the results of the in-sample fit comparison. It shows a close alignment between the model's

Table E.4: Supply Estimation: Only New Titles

	Estimates	SE
Bargaining Parameters *b*		
"Big Five"	0.882	0.114
Smaller Studios	0.647	0.222
Studio Payoff Parameters		
Logged Viewership γ	0.511	0.181
STD of Unobserved Preferences σ_ν	0.094	0.069
Internalization μ	0.379	0.230

Notes. Studios' payoffs are measured in millions of dollars. Standard errors are computed using 100 bootstrap samples.

predictions and actual data on subscription prices, consumer demand, and distribution networks, though it overestimates Amazon's shares in both content licensing and consumer demand while underestimating Hulu's.

Appendix F Additional Counterfactual: Complete Network

In this section, I investigate another counterfactual, where studios are required to license their titles to all of Netflix, Amazon Prime, and Hulu, ensuring a complete distribution network for each title. This counterfactual quantifies the effects of removing all exclusive distribution networks, including those not facilitated by exclusive contracts. Following Section 8.1, I assume the current array of streaming services and titles remains unchanged.

I present the results in the last column of Table E.5. Under the complete network scenario, the downstream competition intensifies significantly, since streaming services can only rely on their in-house content for differentiation. Therefore, comparing the status quo to the counterfactual, all studios and streaming services gain from exclusive contracts, which differs from the findings in Section 8.1.

However, the relative pattern between firms remains consistent. Small studios enjoy additional benefit from exclusive contracts compared to the "Big Five" (+36.2% vs. +3.3%) due to the reliance on using exclusive contracts as a bargaining tool. Small service like Hulu gain substantially more than streaming giants like Netflix and Amazon Prime because they rely more on exclusive third-party content for differentiation. Consumers, however, continue to lose due to reduced title distribution and increased subscription prices, as discussed in Section 8.1.

Table E.5: In-Sample Model Fit & Full Network Counterfactual

		Observed	Simulated Status Quo	Complete Network
Distribution Networks	Netflix	0.712	0.755	1.000
(%Third-Party Titles on)	Amazon Prime	0.185	0.229	1.000
	Hulu	0.248	0.222	1.000
	Only One Service	0.867	0.829	0.000
Subscription Prices	Netflix	13.532	14.678	7.416
(Average $ Per Month)	Amazon Prime	8.990	9.546	8.359
	Hulu	8.407	8.204	6.116
	Disney Plus	7.907	7.829	8.064
Market Shares	Netflix	0.570	0.556	0.613
(Average Per Month)	Amazon Prime	0.461	0.443	0.442
	Hulu	0.358	0.289	0.231
	Disney Plus	0.300	0.299	0.301
	Multi-Homing	0.447	0.414	0.294
Service Payoffs	Netflix		5.940	2.084
($Bn Per Year)	Amazon Prime		2.416	0.570
	Hulu		0.331	0.016
	Disney Plus		2.056	2.174
	Total Service Payoff		10.744	4.844
Studio Payoffs	Big Five		8.470	8.203
($Bn Per Year)	Remaining		4.361	3.201
	Total Studio Payoff		12.831	11.405
Consumer Surplus ($ Per HH-Year)			204.476	329.121
Avg. Weekly Streaming Hours			2.537	3.128

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